Epidemiological Applications of Phylogenetics

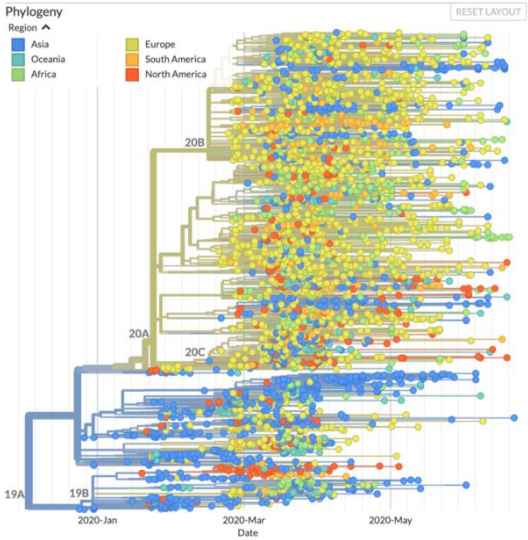
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SLIDES: https://github.com/carolinecolijn/VIASM-lecture

Part 1: Introduction

This session will cover:



- **Introduction**: the relationship between phylogenetics and infectious disease epidemiology
- **Phylodynamics**: Phylogenetics to estimate parameters for ho an infection spreads
- **Phylogeography**: Phylogenetics to reconstruct where a pathogen was in the past
- **Transmission reconstruction**: Phylogenetics to reconstruct who infected whom
- **Discussion and future challenges**

Phylogenetics and epidemiology of infectious disease

- Epidemiological applications of phylogenetics: the use of pathogen sequences to understand infectious disease transmission and dynamics.
- This sub-field is sometimes called *genomic epidemiology*
- “Transmission and dynamics” can mean:
  - pathogen origin (what are the likely sources),
  - time or origin, and for how long has it been undetected
  - infection population dynamics and epidemiological parameters
  - geographic movements of the pathogen
  - person-to-person transmission - who infected or did not infect whom
- Sequence data offer a high-resolution view of how pathogens spread, but the resolution is not perfect.

Whole Genome Sequences

Suppose for sampled pathogens we have a sequence of the whole genome.

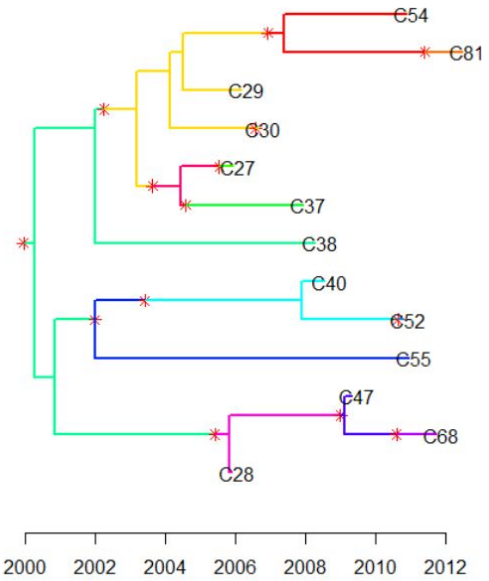
We can use this to estimate how closely related people's infections are.

Challenge: Use this information to make inferences about transmission

The density of sampling shapes what kinds of questions we can answer with genomic data.

Sequence 1: A A C T A G G T A G

Sequence 2: A G T T A G G C G G

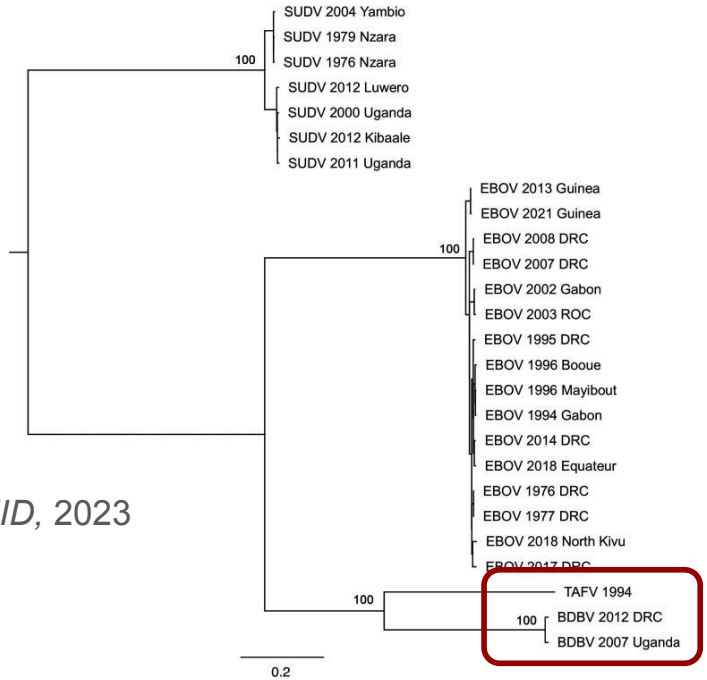


Questions phylogenetics helps answer nearly immediately

- What type or sub-type is this pathogen?
- How long has it been circulating (in humans)?
- What are its nearest known relatives?
- Is it likely to have been a new introduction from an animal population (a *zoonotic* introduction)?
- Or is it likely to be the result of (undetected) ongoing transmission within humans?

EXAMPLE: Ebola outbreak ongoing in the Democratic Republic of Congo

Bundibugyo Ebola virus outbreak



Judson et al, *JID*, 2023

A1: The 2025 DRC outbreak is due to the **Bundibugyo strain**

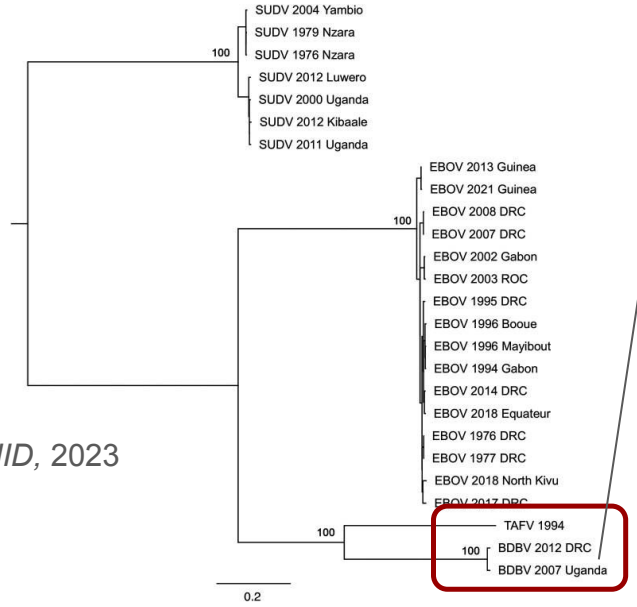
Ebola: Caused by viruses in the *Ortheobolavirus* genus

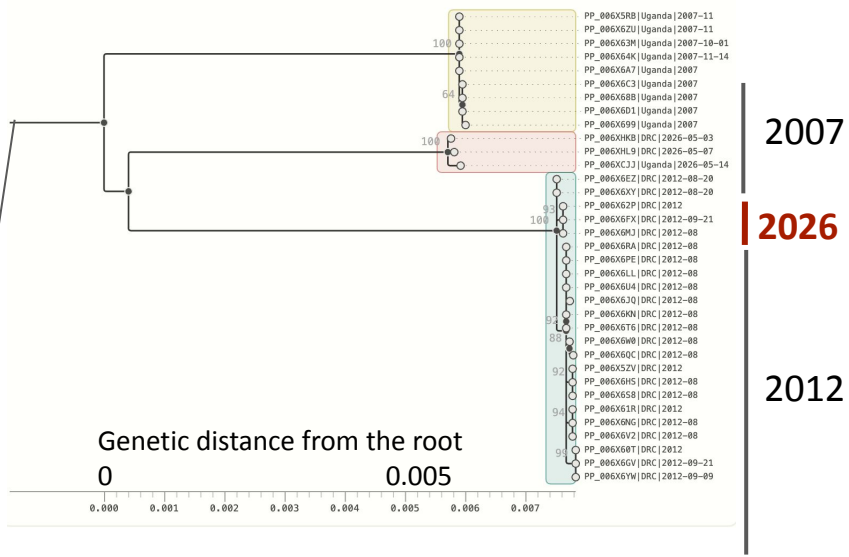
- Of the 6 viruses in the genus, only 4 cause Ebola in humans:
  - Bundibugyo virus
  - Sudan virus
  - Taï Forest virus
  - Ebola virus (formerly Zaire Ebola virus)
- Close relative of Marburg viruses
- Single-stranded negative-sense RNA virus
- Mostly does not spread human-to-human but persists in animal reservoirs (not entirely known which ones)
- Causes severe hemorrhagic fever, death (50% CFR)
- Spreads via bodily fluids, close contact, including at funerals (from the deceased)
- Infection lasts 10-20 days including a latent period

Placing our outbreak in the phylogeny tells us about its origin

Ebola virus overall phylogeny; BDBV zoom in

First 3 sequences
virological.org May 18 2026
Shown with 34 other BDBV sequences





Judson et al, *JID*, 2023

A2: The 2026 cases are likely due to a **new introduction** from an animal reservoir population

A3: Nearest relatives: BDBV outbreaks; Uganda 2007 and DRC 2012

Timed phylogenetic trees

A timed phylogenetic tree has branches in units of calendar time.

These are constructed by combining phylogenetic reconstruction with the assumption of a *molecular clock:* at what rate do substitutions occur, in time? $d = \mu t$

The clock rate μ can best estimated with regression, using sequences gathered over time

Given a clock rate estimate, tools like Least Squares Dating can optimize the tree's branch lengths, giving the best fit to $d = \mu t$, scaling lengths to be in calendar time

This is also done simultaneously in Bayesian phylogenetics (which we will come to)

$$P(\text{tree, times, } \mu \mid \text{sequences}) \propto \mathbf{P(sequences \mid tree, w, \mu)\ P(tree \mid times, \theta)}\ P(\mu)\ P(w)\ P(\theta)$$

phylo likelihood **connection to epidemiology** clock

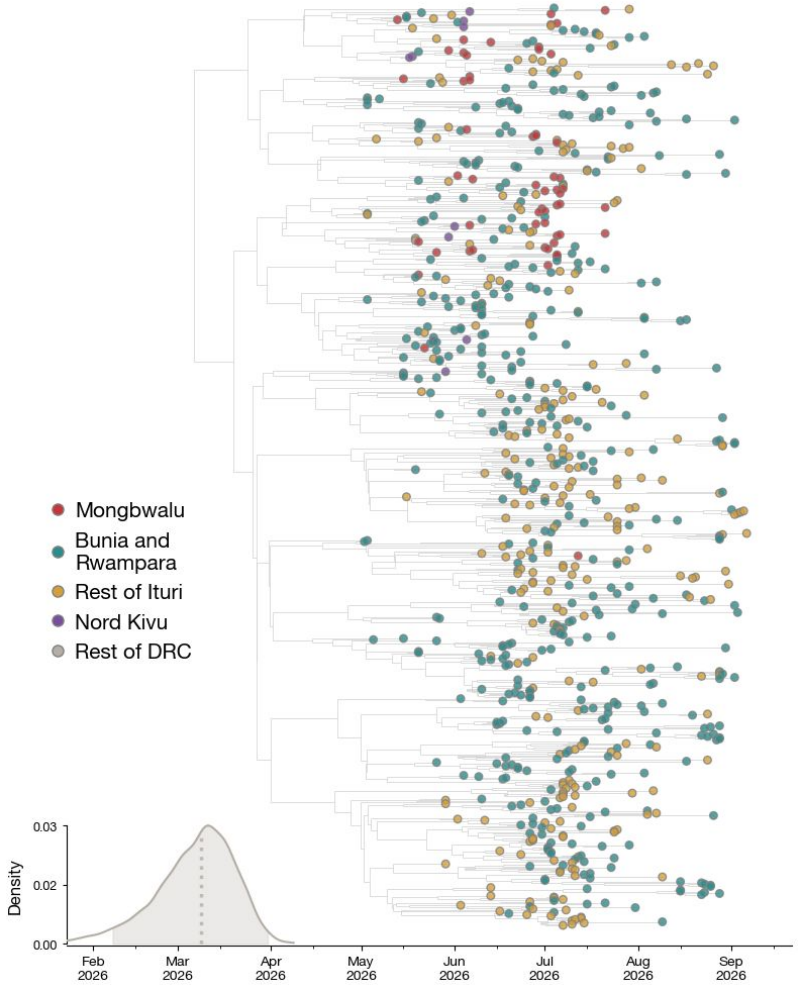
Example: DRC Ebola outbreak as of

Right: Timed phylogenetic tree.

Tips are coloured by location.

Q4: When did this outbreak likely begin?

A4: The tree indicates the likely time of origin: early March, 2026; there is some uncertainty.

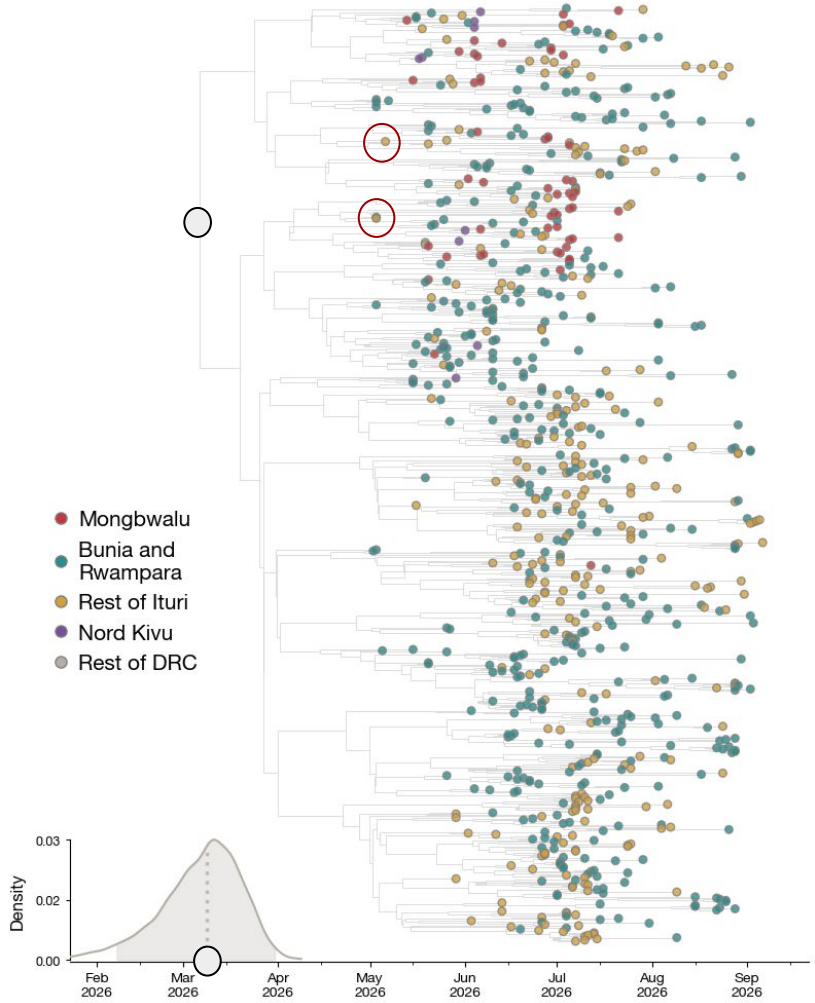


Source: virological.org [Post on October 12, 2026]

Quiz: Timed phylogenetic tree

Consider the two circled cases. They were found in the same region at the same time.

1. How much time elapsed between their most recent common ancestor and the time they were detected?
2. Do you think that their virus genomes are particularly closely related (within the outbreak?)
3. How many transmissions likely separate them?



Quiz Answers

1. How much time elapsed between the circled cases' most recent common ancestor and the time they were detected?

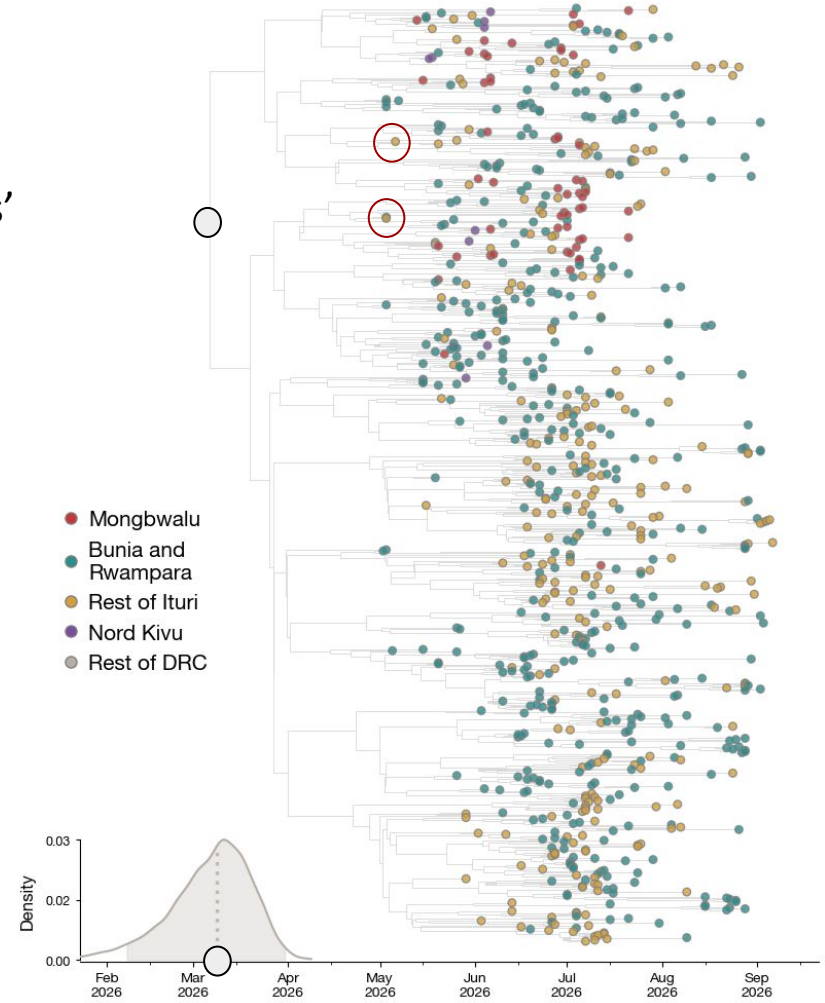
   A: *2 months.*

2. Do you think that their virus genomes are particularly closely related (within the outbreak?)

   A: *No, they are not likely to be very closely related*

3. How many transmissions likely separate them?

   A: *4 months: at least 10-12 infections*



Part 2: Phylodynamics

The term *phylodynamics* usually refers to estimating one or more parameters describing the pathogen's population dynamics, using sequence data. Usually (but not always), sequences are collected over time, and we use a Bayesian inference framework with a posterior:

$$P(\text{tree, times, } \mu, \text{ w, } \theta \mid \text{seqs}) \propto \mathbf{P(seqs \mid tree, w, \mu)\ P(tree \mid \theta)}\ P(\mu)\ P(w)\ P(\theta)$$

phylo likelihood **connection to epidemiology** clock

It's worth talking about:

1. Bayesian phylogenetics

2. where ***P(tree | θ)*** comes from: it's the main link to transmission dynamics.

Aside: Bayesian phylogenetics reminder

$P(tree, times, \mu, w, \theta | seqs) \propto P(seqs \mid tree, w, \mu)\, P(tree \mid \theta)\, P(\mu)\, P(w)\, P(\theta)$

- ***P(sequences | tree, w, μ)****:* computed by Felsenstein's pruning under a molecular evolution model (HKY, JC69, etc) with molecular evolution model *w* and branch lengths ***μ t.***
- ***P(tree | θ):*** the tree prior (this links to the epidemiology). Coalescent or birth-death model gives the probability of the tree shape and node times given the epidemic process
- *P(μ) P(w) P(θ)* priors on the clock rate, evolutionary model parameters, tree prior parameters.

This posterior is high-dimensional with discrete and continuous elements (tree topology + all node times + μ + substitution parameters + tree prior parameters) — no closed form

Explored by **MCMC**: sample trees and parameters proportional to their posterior probability. Tools: BEAST2, BEASTX, MrBayes.

Output: a posterior sample of trees → uncertainty in topology, dates, and parameters is propagated through to everything downstream

Phylodynamics: infer epidemiological parameters

The field of *phylodynamics* links a model with some parameters to the likelihood of a phylogenetic tree.

Then, either

- maximize the likelihood to find the best parameters, given a fixed phylogenetic tree created with data
- use that likelihood as a prior in the Bayesian decomposition to sample θ

$$P(\text{tree, times, } \mu, \text{ w, } \theta \mid \text{seqs}) \propto \mathbf{L(sequences \mid tree, w, \mu)}\ \mathbf{L(tree \mid \theta)}\ P(\mu)\ P(w)\ P(\theta)$$

The Bayesian route is more common, partly for historical reasons, and partly because at outbreak time scales, there can be a lot of uncertainty in the phylogeny.

The main scientific ingredient: *P(tree* | θ*)*

This is either the likelihood, if the tree is fixed, or the "tree prior", in Bayesian methods.

There are two major bodies of work developing P(tree | times, θ):

Coalescent models: condition on the sample size *n* (which should be << the population size *N*). Use an effective population size, *Ne* or *Ne(t)*, with the generation time $\boldsymbol{\tau}$, to write P(tree | times, θ).

Connects to population genetics, Wright-Fisher models.

Birth-death-sampling models: the tree is generated by a birth-death process with birth rate λ, death rate μ and sampling rate ψ (and sometimes other things)

Differences between birth-death and coalescent models

Coalescent: The likelihood is *L*(Tree | sampling times, *Ne(t)*)

The trajectory of Ne(t) does not affect how many samples there are, just the rates at which lineages coalesce.

Birth-death: The likelihood is *L*(Tree, sampling times | birth, death, sampling rate)

The birth, death and sampling rates *do* affect how many samples there are. The more cases you have, the more you are likely to sample.

Suppose you have lots of short branches and closely-related sequences:

- Coalescent: low Ne, this is nothing to worry about.
- Birth-death: this shows fast transmission, worry about this.

The tree likelihood for a birth-death-sampling model: estimating R from sequence data. R: mean number of new infections per infection.

Consider the stochastic process, beginning with one individual:

- Individuals give rise to new individuals at a birth rate λ
- Individuals are removed at a rate μ (recovery or death)
- Individuals are sampled at rate ψ (and this also removes them)

The process proceeds forward in time, and generates a tree.

In this tree, branching events correspond to transmission events. But not all transmission events are in the tree.

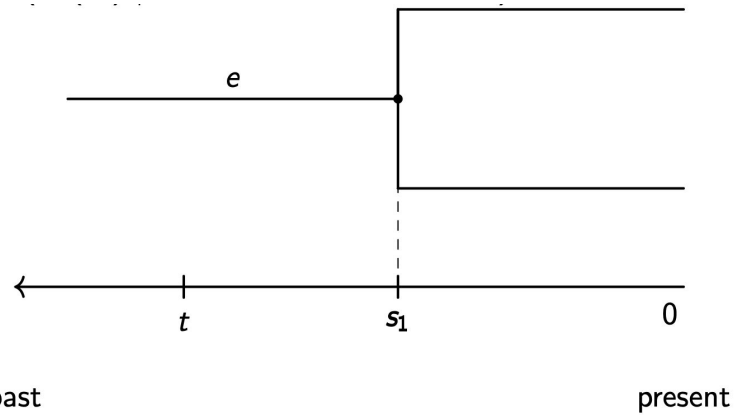
In this process, the *basic reproduction number* is $\lambda/(\mu + \psi)$.

We will build two key quantities: *g* and *p0*

$\mathcal{T}_e(s_1)$: the sub-tree descending from edge e at time s_1

$$p_0(t) = \mathcal{P}\left(\begin{array}{c} \text{a lineage has no sampled} \\ \text{or extant descendants} \end{array} \middle| \text{ the lineage is alive at time } t \right)$$

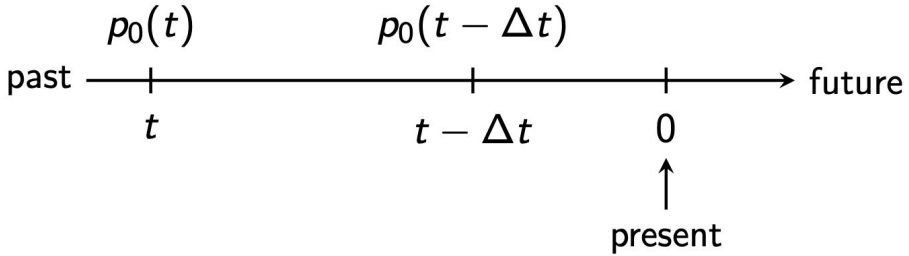
$$g_e(t) = \mathcal{P}(\mathcal{T}_e(s_1) \mid \text{lineage } e \text{ alive at time } t)$$



Differential equation approach

Because the birth-death process is Markovian, we can condition on what happens in the next small interval Δt and take the limit. This is a standard backward equation for a continuous-time Markov chain

Time convention: the present is time *t=0.*
Time increases into the past



Derivation of $p_0(t)$

We use the total law of probability to relate $p_0(t)$ to $p_0(t-\Delta t)$:

$$\underbrace{p_0(t)}_{P(\text{dead by end}|\text{alive at } t)} = \underbrace{\mu\Delta t}_{P(\text{death})} \cdot \underbrace{1}_{P(\text{dead by end}|\text{just died})} + \underbrace{\lambda\Delta t}_{P(\text{birth})} \cdot \left(\underbrace{p_0(t-\Delta t)}_{P(\text{dead by end}|\text{still alive})} \right)^2 + \underbrace{(1-(\mu+\lambda+\psi)\Delta t)}_{P(\text{nothing happens})} \cdot \underbrace{p_0(t-\Delta t)}_{P(\text{dead by end}|\text{still alive})}$$

$$p_0(t) = \mu\Delta t \times 1 + \lambda\Delta t p_0(t-\Delta t)^2 + (1-(\mu+\lambda+\psi)\Delta t)p_0(t-\Delta t),$$

$$\Longrightarrow$$

$$p_0(t) - p_0(t-\Delta t) = \left(\mu + \lambda p_0(t-\Delta t)^2 - (\mu+\lambda+\psi)p_0(t-\Delta t)\right)\Delta t,$$

$$\Longrightarrow$$

$$\frac{p_0(t) - p_0(t-\Delta t)}{\Delta t} = \left(\mu + \lambda p_0(t-\Delta t)^2 - (\mu+\lambda+\psi)p_0(t-\Delta t)\right).$$

This gives a differential equation for $p_0(t)$, the probability of having no descendants at the present time if alive at time t.

$$\frac{dp_0(t)}{dt} = \mu + \lambda p_0(t)^2 - (\lambda + \mu + \psi)p_0(t)$$

For a tip sampled at time $s > 0$, the initial condition for p_0 is

$$p_0(0) = 1$$

because sampling means being removed. There can be no descendant *at the present time* $t = 0$.

The DE for $p_0(t)$ does not depend on the tree, just the parameters. It is solved explicitly, then used in the next part.

$$p_0(t) = \frac{\lambda + \mu + \psi + c_1 \frac{e^{-c_1 t}(1-c_2)-(1+c_2)}{e^{-c_1 t}(1-c_2)+(1+c_2)}}{2\lambda}$$

solution in Stadler 2010

We do the same for the quantity g

$$g_e(t) = \mathcal{P}(\mathcal{T}_e(s_1) \mid \text{lineage } e \text{ alive at time } t)$$

$$g_e(t+\Delta t) = (1-(\lambda+\mu+\psi)\Delta t - O(\Delta t^2))g_e(t) + \lambda \Delta t\, 2p_0(t)g_e(t) + O(\Delta t^2)$$

The first term: nothing happens.

The second term: There is a branching event, one side dies off ($p_0(t)$) and the other side produces the tree $\mathcal{T}_e(s_1)$. There are two ways this can happen (two descendants).

Taking $g_e(t+\Delta t) - g_e(t))$ and taking the limit as $\Delta t \to 0$ gives the differential equation for $g_e(t)$:

$$\frac{dg_e(t)}{dt} = -(\lambda+\mu+\psi)g_e(t) + 2\lambda p_0(t)g_e(t).$$

Initial conditions for g

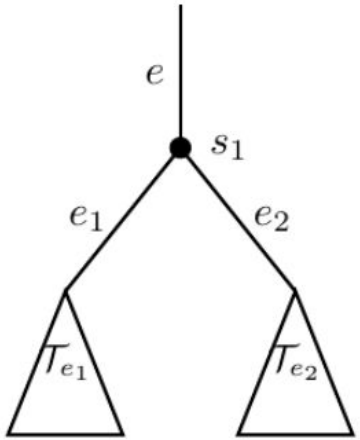
We still need initial conditions for the equations for $g_e(t)$.

Edge e ends either in a sampling event or in a branching event. Suppose e ends at time s_1. Set

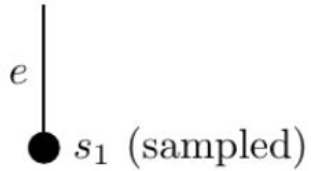
$$g_e(s_1) = \begin{cases} \lambda g_{e_1}(s_1) g_{e_2}(s_1) & \text{if } e \text{ has two descendant edges } e_1, e_2, \\ \psi & \text{if } e \text{ has no descendant edges.} \end{cases}$$

Up to this point $g_e(t)$ has been defined as a probability, but because sampling times are observed continuous random variables, the initialisation $g_e(s) = \psi$ makes g_e a density with respect to those times — so from here we treat it as a likelihood contribution rather than a probability. The ODE and its structure are unchanged; only the interpretation shifts at the tips.

Two cases for the initial conditions for *g*



$$g_e(s_1) = \lambda\, g_{e_1}(s_1)\, g_{e_2}(s_1)$$



$$g_e(s_1) = \psi$$

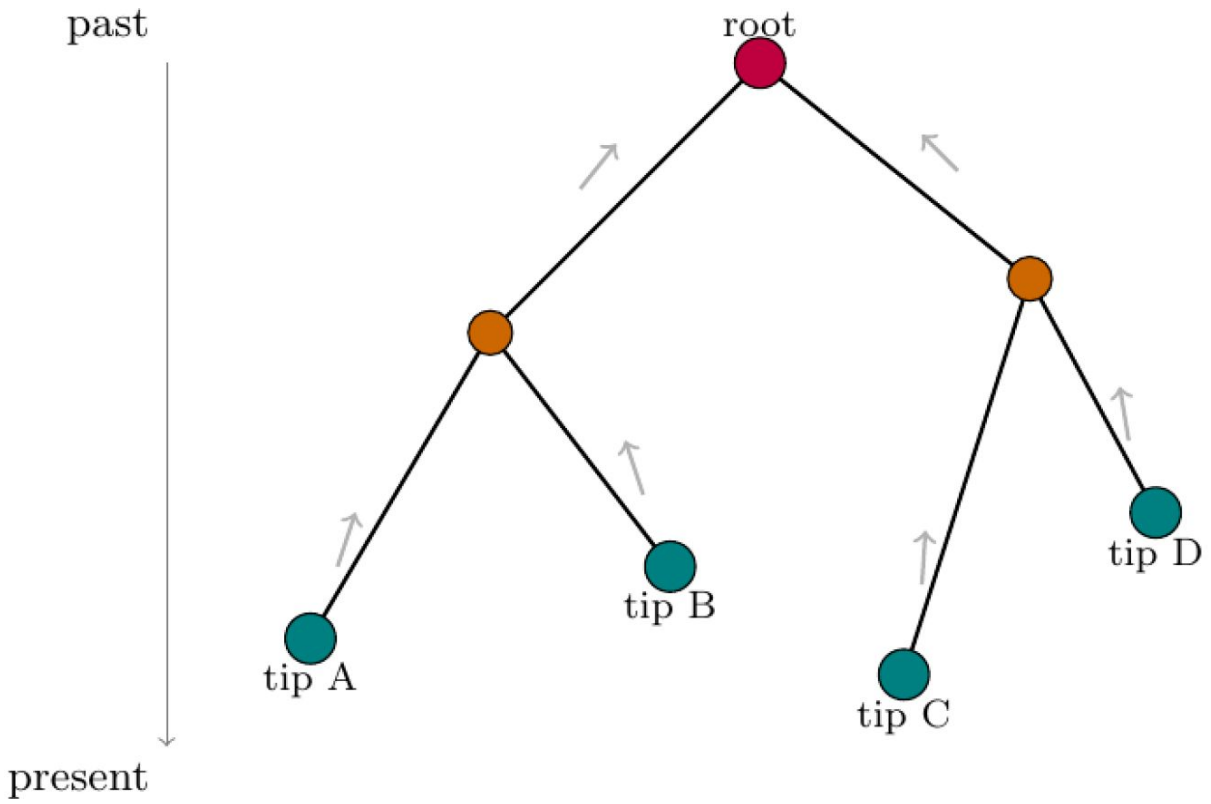
Stadler (2010) give the explicit solution to the ODE for *g*.

Building the tree likelihood for this birth-death-sampling model

- Solve the ODE for $p_0(t)$.
- Beginning at the tips, initialize $g_e(t)$ and solve the ODE for g_e, using the $p_0(t)$ solution, along the edges. You then obtain g_e for all terminal branches, at the times they originate.
- Continue working up the tree, setting $g_e(s_1) = \lambda g_{e_1}(s_1) g_{e_2}(s_1)$ at each branching event (joining the g solutions for edges e_1 and e_2 to start the equation for g on edge e
- When you reach the root of the tree, you have the tree likelihood!

$$L(\mathcal{T}|\lambda, \mu, \psi) = g_r(t_0)$$

where r is the root and t_0 is its time.



Step 1: solve $p_0(t)$ everywhere on tree

Step 2: tips $g_e(s) = \psi$ solve ODE upward

Step 3: at each node $g_e(s_1) = \lambda\, g_{e_1}(s_1) g_{e_2}(s_1)$ solve ODE upward

Step 4: root $\Rightarrow$ likelihood!

Now that you have $L(T|\theta)$ ..

For a fixed phylogenetic tree $\mathcal{T}$ you could find the parameters λ, μ and ψ for which the likelihood $g_r(t_0)$ is highest.

But $\mathcal{T}$ is not known completely. And there are some identifiability issues: the three parameters are not all identifiable.

The field uses Bayesian inference to jointly sample $\mathcal{T}$ with λ, μ and ψ. We can use prior knowledge, for example we probably know how long an infection lasts. This informs μ.

BEAST X, BEAST2: Propose a tree. Compute its likelihood. Compute the priors. Combine. Accept or reject. Continue. Result: a sample of trees and birth-death-sampling parameters.

Epidemiological parameters from phylogenetics

There are many models that can give a likelihood $L(T \mid \Theta)$.

Birth-death sampling variations include:

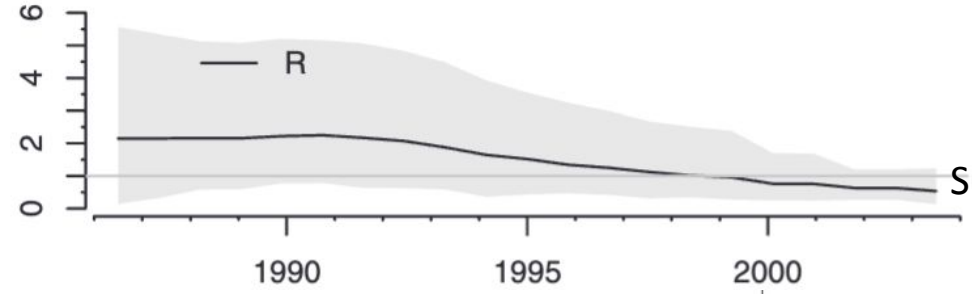
- “Sampled ancestors” (sampling does not remove the lineage - e.g. fossils)
- Multiple types, with different birth and death rates - like variants of concern
- Variable birth and death rates -- allows inference of reproduction number through time.

Coalescent variations include:

- Varying the effective population size through time, *Ne(t).*
- Epidemiological coalescent: susceptible-infectious-recovered (SIR) models, SEIR, etc
- Both incidence (new cases) *and* prevalence (*N*) contribute to coalescent rates
- “Structured populations” - lineages more likely to coalesce with others in the same “deme” (sub-population, such as a geographic region)

Application: R(t) in HIV and Hepatitis C (Stac 2012)

HIV-1 subtype B in the United Kingdom; 62 sequences 1999-2003. R declines consistent with the development of multi-drug therapy.



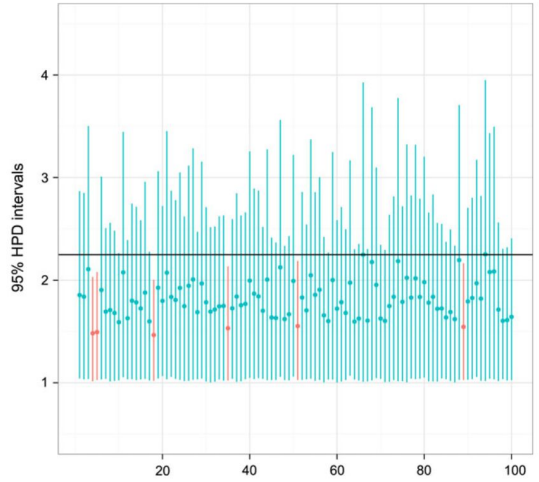
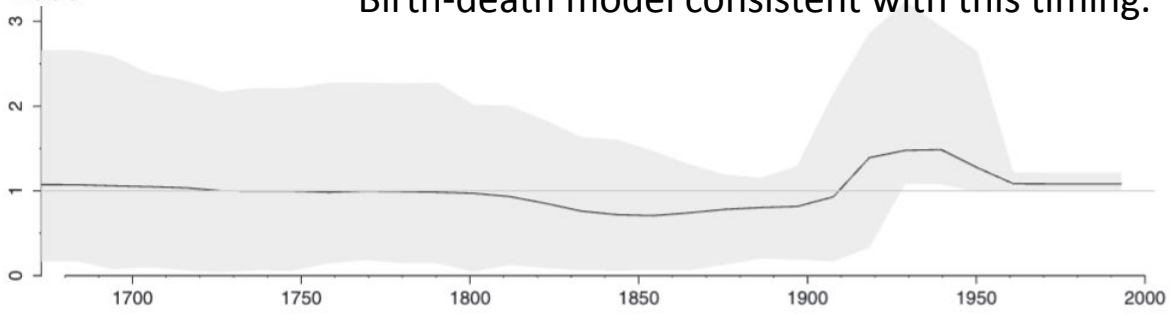


FIG. 4. The estimated mean R_0 together with the 95% HPD for the 100 simulated trees (true $R_0 = 2.29$).

HCV in Egypt: 63 sequences.
Suspected cause of increase: contaminated injections against bilharziosis.
Birth-death model consistent with this timing.



Quiz: phylodynamics

1. What is the main purpose of phylodynamics?
2. What are the two main sets of models used to set up tree likelihoods for phylodynamics?
3. What is the most common inference approach?

Quiz Answers

1. What is the main purpose of phylodynamics?

   A: *To estimate one or more parameters about the infection's dynamics using sequence data (usually phylogenetics)*

2. What are the two main sets of models used to set up tree likelihoods for phylodynamics?

   A: *Coalescent theory and birth-death models*

3. What is the most common inference approach?

   A: Bayesian inference. The phylogenetic tree itself is not certain. However, deep learning and maximum likelihood methods are also used.

Limitations

B-D models tend to underestimate R0.

This math was complicated but the birth-death-sampling model is very simple: perpetual exponential growth is not realistic. Population structure; fewer susceptibles as the outbreak progresses; super-spreading; behavioural responses; contact tracing..

Alternative: Deep learning!

- Simulate many trees under a model
- Encode the trees.
- Train a ML to infer the parameters using the simulations as a ground truth. No likelihood needed!

JOURNAL ARTICLE

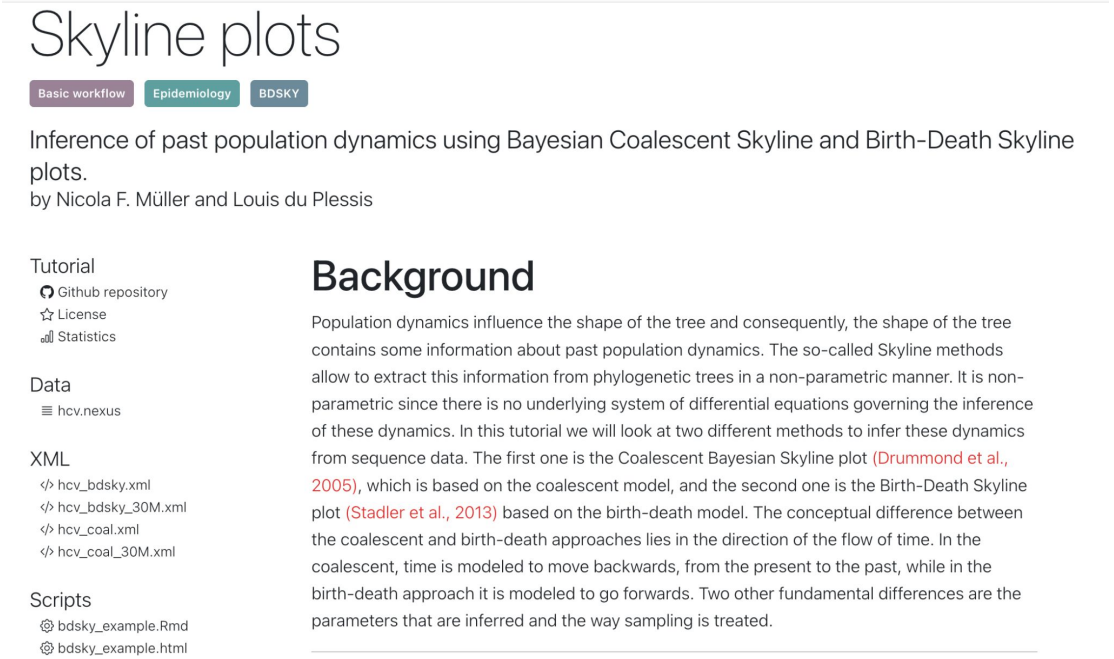
PhyloCNN: Improving Tree Representation and Neural Network Architecture for Deep Learning from Trees in Phylodynamics and Diversification Studies

Manolo Fernandez Perez ✉, Olivier Gascuel ✉

Systematic Biology, Volume 75, Issue 4, July 2026, Pages 776–795,
https://doi.org/10.1093/sysbio/syaf082
Published: 22 November 2025 **Article history** ▾

Phylodynamics and other tutorials at Taming the BEAST

Learn how to use the birth-death and coalescent “skyline” models to estimate how epidemiological parameters (either R or Ne) vary over time



Skyline plots

Basic workflow | Epidemiology | BDSKY

Inference of past population dynamics using Bayesian Coalescent Skyline and Birth-Death Skyline plots.

by Nicola F. Müller and Louis du Plessis

Tutorial
- Github repository
- License
- Statistics

Data
- hcv.nexus

XML
- hcv_bdsky.xml
- hcv_bdsky_30M.xml
- hcv_coal.xml
- hcv_coal_30M.xml

Scripts
- bdsky_example.Rmd
- bdsky_example.html

Background

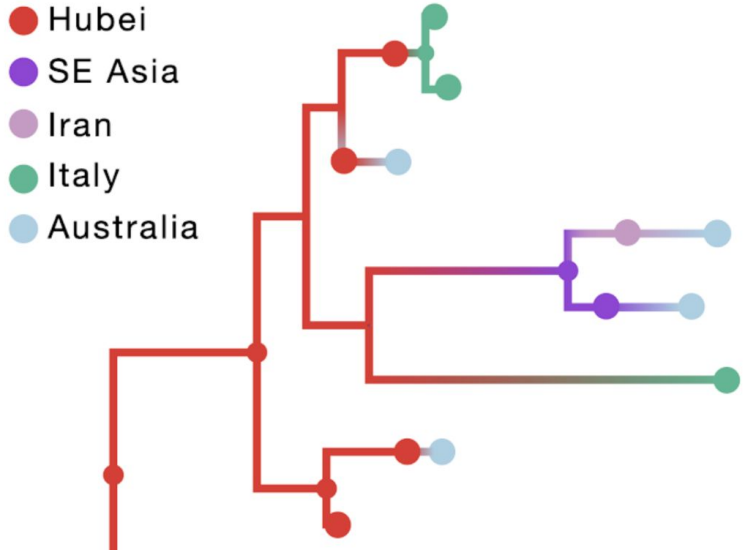
Population dynamics influence the shape of the tree and consequently, the shape of the tree contains some information about past population dynamics. The so-called Skyline methods allow to extract this information from phylogenetic trees in a non-parametric manner. It is non-parametric since there is no underlying system of differential equations governing the inference of these dynamics. In this tutorial we will look at two different methods to infer these dynamics from sequence data. The first one is the Coalescent Bayesian Skyline plot (Drummond et al., 2005), which is based on the coalescent model, and the second one is the Birth-Death Skyline plot (Stadler et al., 2013) based on the birth-death model. The conceptual difference between the coalescent and birth-death approaches lies in the direction of the flow of time. In the coalescent, time is modeled to move backwards, from the present to the past, while in the birth-death approach it is modeled to go forwards. Two other fundamental differences are the parameters that are inferred and the way sampling is treated.

https://taming-the-beast.org/tutorials/Skyline-plots/

Part 3: Phylogeography - estimating where pathogens were in the past

"Where": can be geographic location, or even host species.

Location can be discrete or continuous. Here: discrete state only.



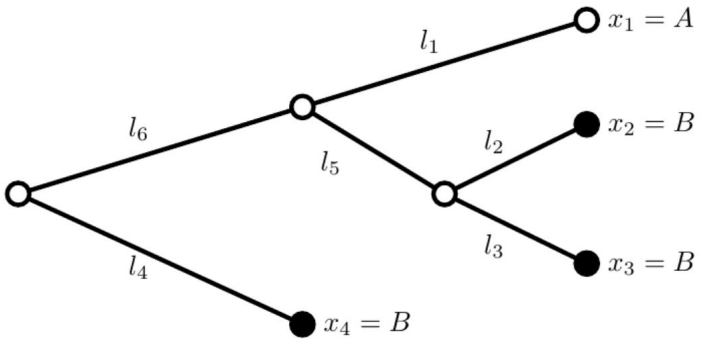
For much more on phylogeography, see Alex Beams' week-long workshop at: https://alexbeams.github.io/phylogeography_workshop/

Part 3a: Classic discrete state phylogeography
A two-location example illustrating the question

Here is a tree with 4 tips: 1, 2, 3, 4.

There are two locations: A (white) and B (black).

The question for discrete trait phylogeography is: where was the pathogen at the times of the internal nodes? (Phylogeography can also ask: where was the pathogen at any arbitrary point on an edge. But we will talk about only the internal nodes)



Key assumption: location can be modelled as evolving on the tree, just like a character in the DNA or RNA sequence. **Inference: just like for molecular evolution.**

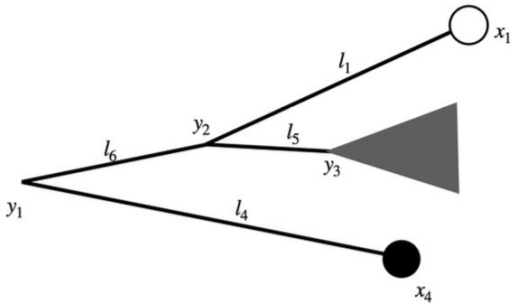
Location as a Markov process on the tree (like RNA or DNA)

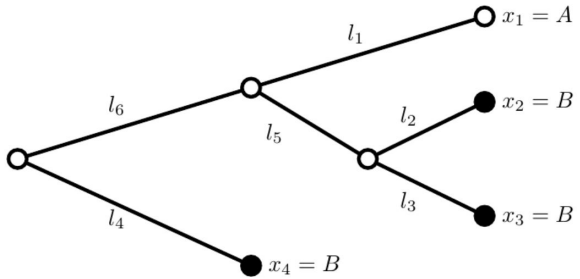
- The location of the pathogen at any point on the tree evolves as a continuous-time Markov chain with state space = set of locations $\{1, 2, \ldots, k\}$
- The rate matrix Q has entries q_{ij} = rate of moving from location i to location j, with $q_{ii} = -\sum_{j \neq i} q_{ij}$
- The transition probability matrix over a branch of length ℓ is $P(l) = e^{Ql}$, so $P_{ij}(\ell)$ = probability of being in location j at the end of a branch of length ℓ, given starting in location i
- Q is estimated from the data. The elements q_{ij} are the instantaneous rates of transition between states i and j. (Usually assumed to be symmetric).
- The likelihood is computed by Felsenstein pruning, exactly as for nucleotide substitution. Tip states are known. At each internal node, we sum over all possible ancestral locations
- The stationary distribution π of Q gives the long-run probability of being in each location, and we use that for the probability for the state of the root.

Recall the Felsenstein pruning algorithm

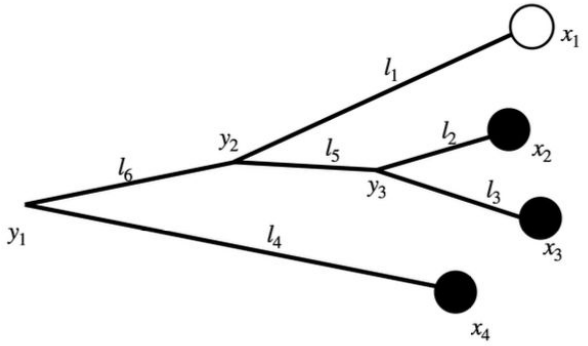
The notation $L_{y_k}(z)$ refers to the likelihood of the tree descending from node y_k, given that y_k has state z.

So we have as many terms $L_{y_3}(z)$ as there are possible values, z, for the node y_3





Recall Felsenstein's pruning (2)



$$L_{y_3}(z) = p(x_2 \mid y_3 = z, l_2)\, p(x_3 \mid y_3 = z, l_3)$$

$$L_{y_2}(z) = p(x_1 \mid y_2 = z, l_1) \left(\sum_w p(y_3 = w \mid y_2 = z, l_5)\, L_{y_3}(w) \right)$$

$$L_{y_1}(z) = p(x_4 \mid y_1 = z, l_4) \left(\sum_u p(y_2 = u \mid y_1 = z, l_6)\, L_{y_2}(u) \right)$$

Finally, the likelihood for the whole tree is the combination of likelihoods of the tree given that the root is in state z:

$$L_{y_1} = \sum_z \pi(z) L_{y_1}(z)$$

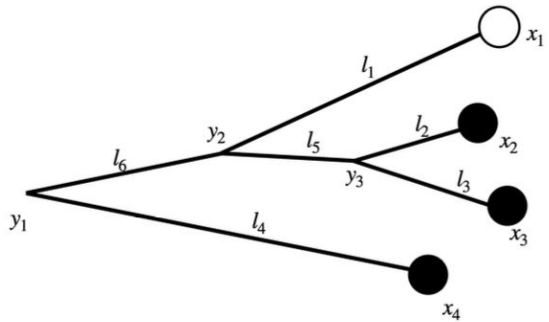
Interpretation for phylogeography

The likelihood we found was the likelihood that the tip states ended up where they did, given the Markov model (rate matrix Q) and the tree:

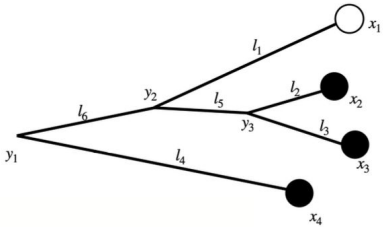
$$P(\mathbf{x} \mid \mathcal{T}) = \sum_{\mathbf{y}} \pi(y_1)p(x_4|y_1)p(y_2|y_1)p(x_1|y_2)p(y_3|y_2)p(x_2|y_3)p(x_3|y_3)$$

This is a **marginalization of a joint distribution**. It has the form

$$p(\mathbf{x} \mid \mathcal{T}) = \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y} \mid \mathcal{T}).$$



The “joint distribution” (of ancestral states **y**)



We seek the set of internal node states **y** that gives the *biggest* contribution to the sum, ie the largest $p(\mathbf{x}, \mathbf{y} \mid \mathcal{T})$.

Hard problem! Solution: while performing Felsenstein’s pruning, *record* the values of $L_{\text{node}}(\text{node state})$ that are computed.

Start by setting $\hat{y}_1 = \text{argmax}_z \pi(z) L_{y_1}(z)$ as the most likely root state.

Then, given $\hat{y}_1$, set $\hat{y}_2 = \text{argmax}_{y_2} p_{\hat{y}_1, z}(l_6) L_{y_2}(z)$. This looks complicated. But it’s just “set y_2 to the state that gives the highest contribution”.

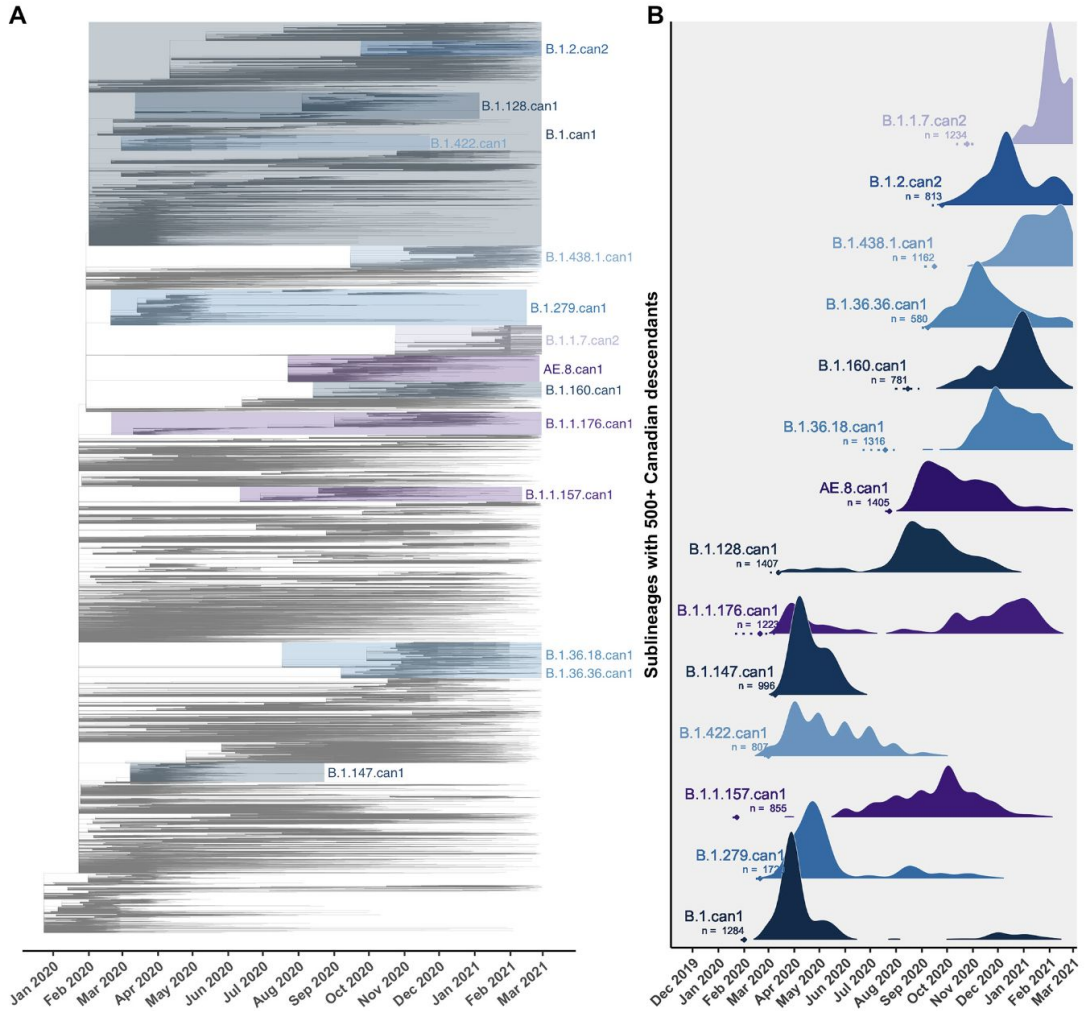
Then, continue down the tree, setting each node’s state to the one that makes the highest contribution to $\sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y} \mid \mathcal{T})$.

COVID-19 application

50,000 sequences including most available Canadian sequences (2021)

Left: phylogenetic tree, coloured by COVID lineage

Right: times of nodes of lineages with > **500 Canadian descendants**. This inference was done with phylogeography (ace).



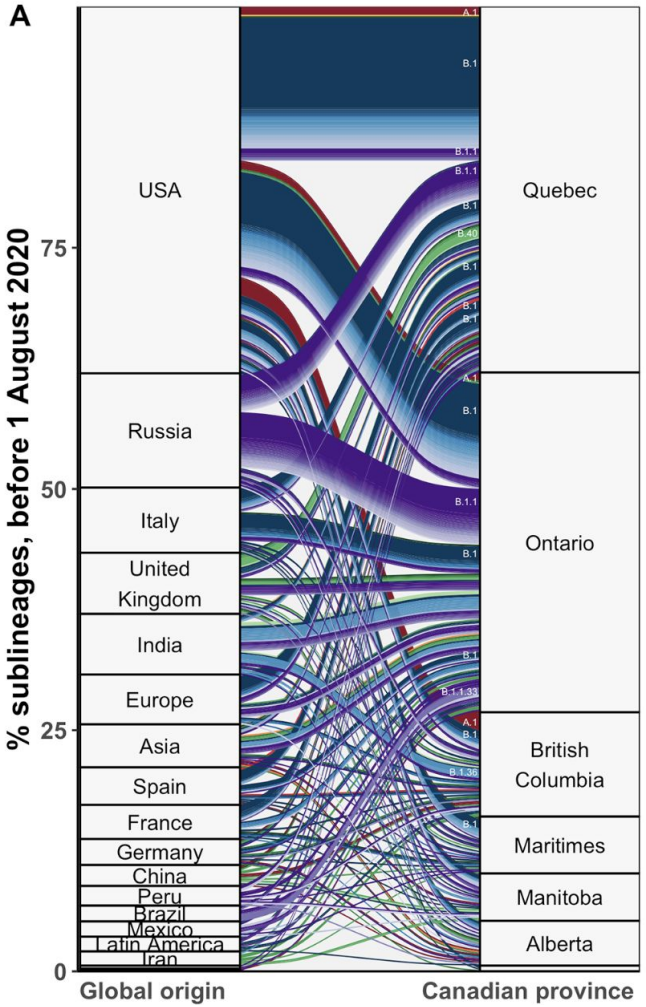
Genomic epidemiology of the first two waves of SARS-CoV-2 i

Angela McLaughlin [1,2,✉], Vincent Montoya [1], Rachel L Miller [1,2], Gideon J Mordecai [3]; Canadian Genomics Network (CanCOGen) Consortium[‡], Michael Worobey [4], Art FY Poon [5], Jeffrey B Joy

COVID-19 movements

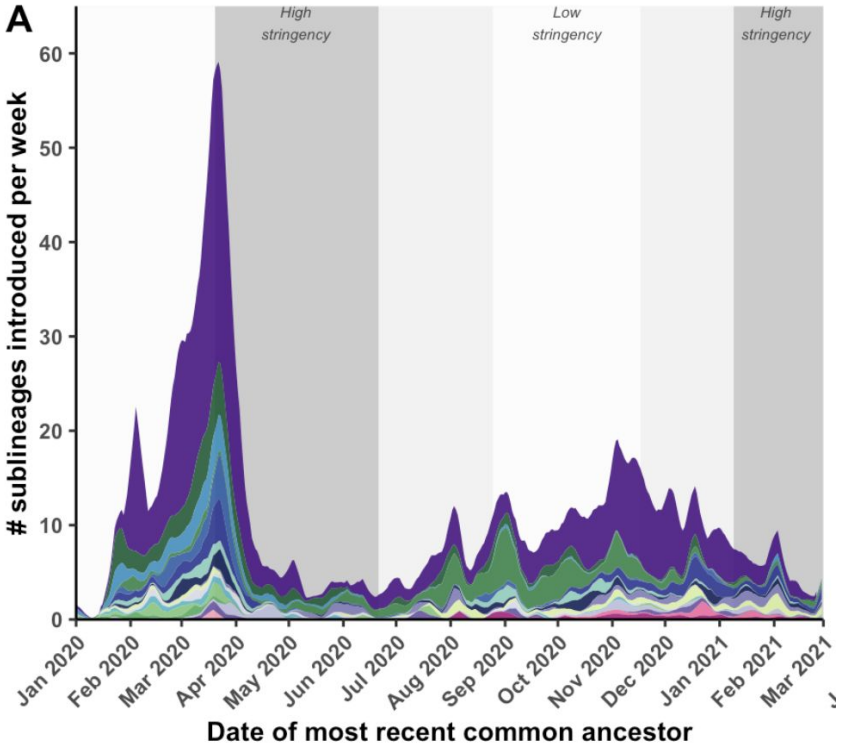
These “alluvial plots” illustrate the inferred viral movements.

They are formed by counting edges with a parent node in one state (eg Russia) and the child node in another (eg Ontario, a province of Canada).



Number of introductions (that are ancestral to the sample)

The authors also asked how many edges were there in the tree with ancestral states **y**, at a given time, with the parent node outside Canada and the child node in Canada?



Bayesian phylogeography: Lemey et al 2009

The tree is not known with certainty.

The Bayesian approach samples a posterior:

$$P(T, Q, \mu \mid \mathbf{x}, D) \propto P(\mathbf{x} \mid T, Q)\, P(D \mid T, \mu)\, P(T)\, P(Q)\, P(\mu)$$

where $\mathbf{x}$ are the observed tip locations, D is the sequence data, T is the tree, Q is the location rate matrix, and μ is the clock rate (and substitution model parameters).

The two likelihoods $P(\mathbf{x} \mid T, Q)$ (which is what we just derived) and $P(D \mid T, \mu)$ are independent given T: the location and sequence CTMCs run on the same tree but are otherwise separate. (Note, this is an assumption!)

The Bayesian approach links the phylogeography to the tree

priors

$$P(T, Q, \mu \mid \mathbf{x}, D) \propto P(\mathbf{x} \mid T, Q)\, P(D \mid T, \mu)\, P(T)\, P(Q)\, P(\mu)$$

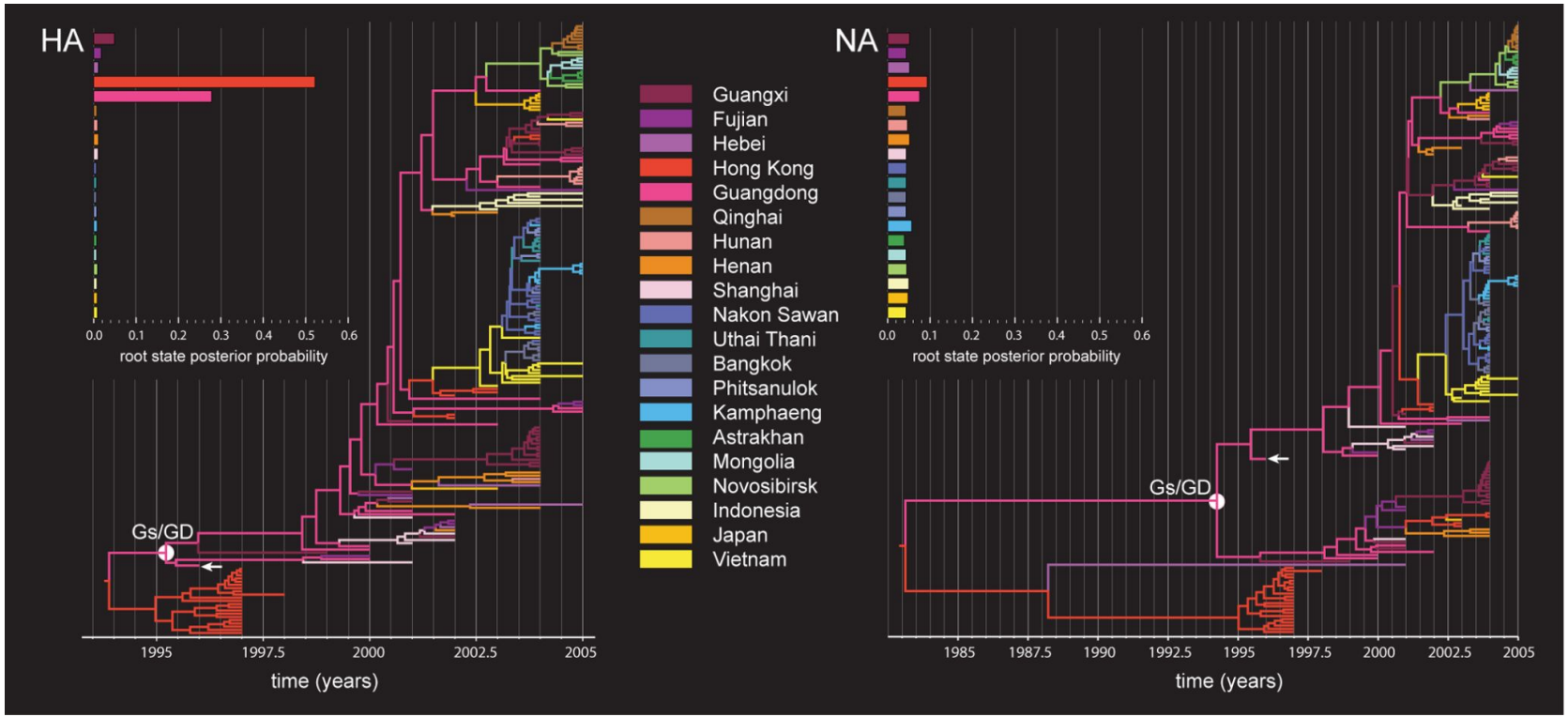
Likelihood for the observed locations given the tree and the phylogeography rate matrix Q

Likelihood for the sequence data given the tree, evolutionary model and clock rate

If we sample from this posterior, the trees on which the location pattern is very unlikely for any plausible Q will not be sampled.

In this way, the discrete trait phylogeography assumption affects which trees we think are more or less likely, and therefore which trees we infer.

Phylogeography of 192 A-H5N1 strains, 20 locations

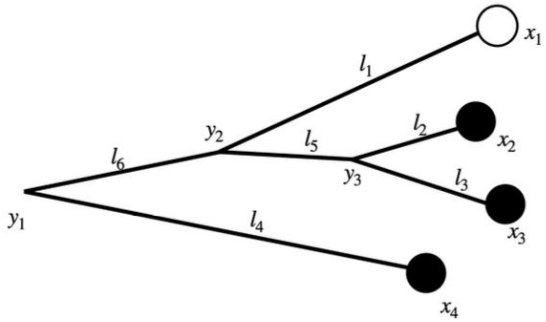


Conclusions about where H5N1 was in the past.
Simultaneous inference of the tree and the other parameters

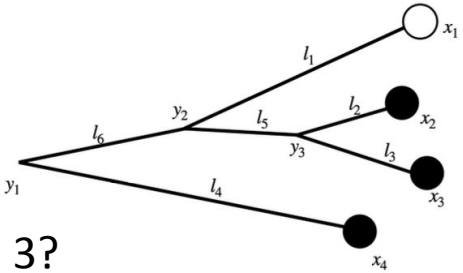
Lemey et al 2009

Quiz: Phylogeography

1. What is the most likely location for the node joining tips 2 and 3?
2. If branch 1 was *much* longer, would you be more or less confident about the state of the node at y2? Why?
3. In the COVID-19 application, is the number of introductions likely to be an under-estimate or an over-estimate? why?



Quiz Answers



1. What is the most likely location for the node joining tips 2 and 3?

   A: *Black. Both descendants are black and relatively near the node*

2. If branch 1 was *much* longer, would you be more or less confident about the state of the node at y2? Why?

   A: *More confident that the state is black. y3 is likely to be black. As we move tips away from the nodes, the information the tip contains about the node state is reduced, so tips 2 and 3 will provide stronger information.*

3. In the COVID-19 application, is the number of introductions likely to be an under-estimate or an over-estimate? why?

   A: *An under-estimate. Only introductions with sampled descendants can appear, and of those, we might miss some if we mis-estimate the node states.*

A problem with discrete trait phylogeography

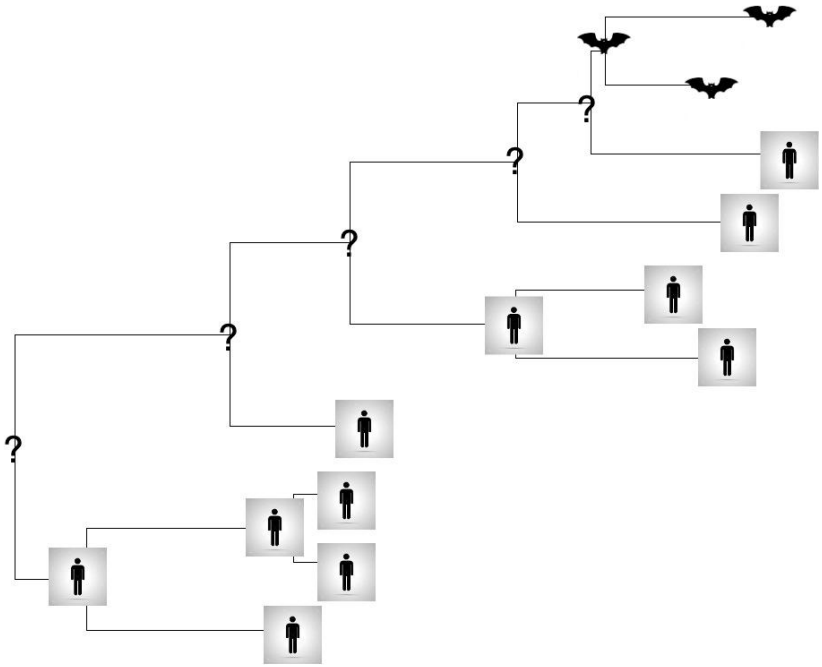
Sampling differences in different states can greatly bias the ancestral state reconstructions.

Note that there is no term related to how likely sampling is, in our computations.

Conceptual example: Ebola virus (right)

The internal nodes: what state are they in? Should they be in bats or humans?

A Markov model like ace will likely place all but the last one in humans. (Why?)



Part 3b: Large-scale phylogeography with adjustment for known sampling differences

- Suppose the UK samples and sequences more of its SARS-CoV-2 infections than the USA.
- Or: the USA sequences more influenza viruses than many countries.
- Large-scale viral phylogeography analyses (10,000s of taxa) and tools like nextstrain use downsampling to adjust for this. (eg the COVID-19 paper we just saw)
- But this is time-consuming and discards a lot of data.

Our approach: SAASI. Nature Comms 2026
Thank you to Yexuan Song.



State-dependent Speciation and Extinction (SSE) models

These models began with a paper by Maddison, Midford and Otto, Systematic Biology, 2007 (BiSSE: Binary SSE) and were not used for phylogeography. Their goal: the branching in the tree depends on the state.

In our application, we specify rates at which a lineage in state $i \in \mathcal{S}$ (e.g. location, host species) does the following things:

Event	Rate
Speciates (transmits), giving daughter states j, k	$\Lambda_{i,j,k}$
Dies (recovers/clears)	μ_i
Is sampled	ψ_i
Transitions from state i to j	q_{ij}

The **underlying stochastic process is a birth-death process**, but with many types (corresponding to state in *S*).

Each state has its own branching, death and sampling rates.

There is a transition matrix between states.

Goal: Ancestral state reconstruction in the SSE model.
Why? Mainly to account for sampling differences between states (but could also account for differences in other parameters)

We compute two key quantities (generalizations of the the birth–death–sampling quantities g_e, p_0):

Let $D_{e,i}(\tau) = P(\mathcal{T}_e(\tau) \mid Y_e(\tau) = i, \theta)$ be the probability that edge e in state i at (backward) time τ gives rise to the *observed* subtree below it. This is like g_e in our birth-death model.

Let $E_i(\tau) = P(\text{no observed descendants} \mid \text{lineage in state } i \text{ at time } \tau)$. This is like $p_0(\tau)$.

Differential equation for the Ds (possible because this is still a Markov process - memoryless)

$D_{e,i}(\tau)$ satisfies (like the ODE for g_e):

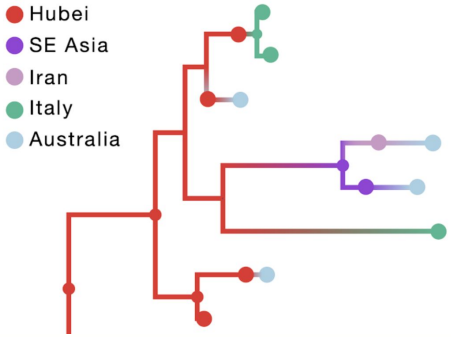
$$\frac{d}{d\tau}D_{e,i} = -\left(\sum_{j,k}\Lambda_{i,j,k} + \mu_i + \psi_i + \sum_{j\neq i} q_{ij}\right) D_{e,i} + \sum_{j,k} 2\Lambda_{i,j,k}\, D_{e,j}\, E_k + \sum_{j\neq i} q_{ij}\, D_{e,j}$$

no speciation, death or sampling occurs — speciation creates a j, k pair; 2 ways to do this — state transition

Initial conditions at a node $\tau_{e,0}$:

$$D_{e,i}(\tau_{e,0}) = \begin{cases} \psi_i & \text{(sampled tip)} \quad \longleftrightarrow \quad g_e(s) = \psi \\ \displaystyle\sum_{j,k} \Lambda_{i,j,k}\, D_{e_1,j}(\tau_{e,0})\, D_{e_2,k}(\tau_{e,0}) & \text{(speciation node)} \quad \longleftrightarrow \quad g_e(s_1) = \lambda\, g_{e_1} g_{e_2} \end{cases}$$

Differential equation for the Es



$E_i(\tau)$ satisfies (like the ODE for p_0):

$$\frac{d}{d\tau}E_i = \mu_i - \left(\sum_{j,k}\Lambda_{i,j,k} + \mu_i + \psi_i + \sum_{j\neq i} q_{ij}\right)E_i + \sum_{j,k} 2\Lambda_{i,j,k}\, E_j\, E_k + \sum_{j\neq i} q_{ij}\, E_j,$$

with $E_i(0) = 1$ (at the present, a lineage has probability 1 of leaving no future observed descendants).

With just one state and no Q (no state transitions) these equations reduce to the previous birth-death-sampling model equations.

Approach to assigning ancestral states

Our goal: get the analogue of the joint distribution p(***x, y*** | *T*) for the internal node states, as we did with classic phylogeography.

Recall what we did before: we chose the set of states **y** that made the biggest contribution to the sum

$$p(\mathbf{x} \mid \mathcal{T}) = \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y} \mid \mathcal{T}).$$

To do that, we used the information we got from Felsenstein's pruning: this uses the likelihood of the **tip states x** we observe *below* a node if it's in state z, combined with how likely the node is to be in state z given what came above it.

We do the same here. D and E tell us the first part. We need the second part.

What came before our edge? Condition on the "complement tree"

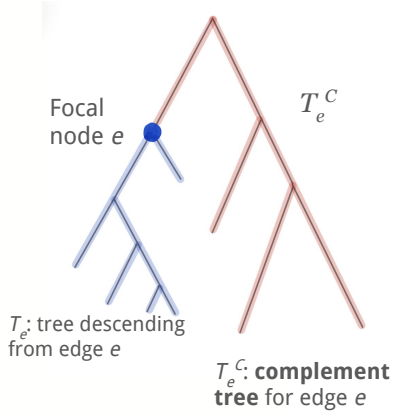
$D_{e,i}$ and E_i have information *below* edge e: how likely is the tree that comes after edge e in state i?

To infer ancestral states we also need information *above* edge e (the "ancestral complement" $\mathcal{T}_e^C$). Define (forward in time $t = T - \tau$):

$$\widetilde{D}_{e,i}(t) = \Pr\big(Y_e(T-t) = i \mid \mathcal{T}_e^C(T-t), \theta\big)$$

For *anagenetic* (state-transition only) changes:

$$\frac{d}{dt}\widetilde{D}_{e,i}(t) = -\sum_{j \neq i} q_{ij}\,\widetilde{D}_{e,i}(t) + \sum_{j \neq i} q_{ji}\,\widetilde{D}_{e,j}(t)$$



These are initialized at each node from the parent edge and sister subtree (see paper for full expression).

Combining the forward and backward information

Ancestral state probability at edge e:

$$A_{e,i}(\tau) = \frac{\widetilde{D}_{e,i}(T-\tau)\, D_{e,i}(\tau)}{\sum_j \widetilde{D}_{e,j}(T-\tau)\, D_{e,j}(\tau)}$$

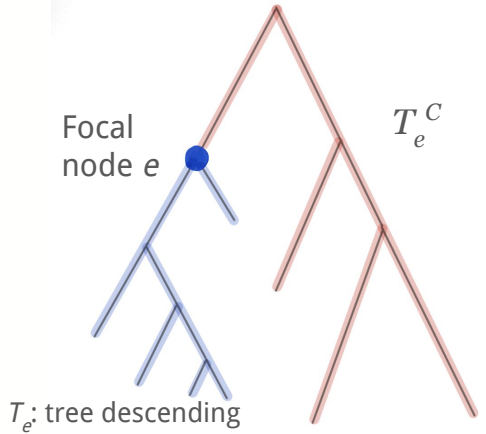
Probability edge is in state i given Tc

Likelihood of tree below edge *e* given that the state of *e* is *i*

The denominator normalises: $\sum_i A_{e,i}(\tau) = 1$.

Key innovation (SAASI): ψ_i differs by state.

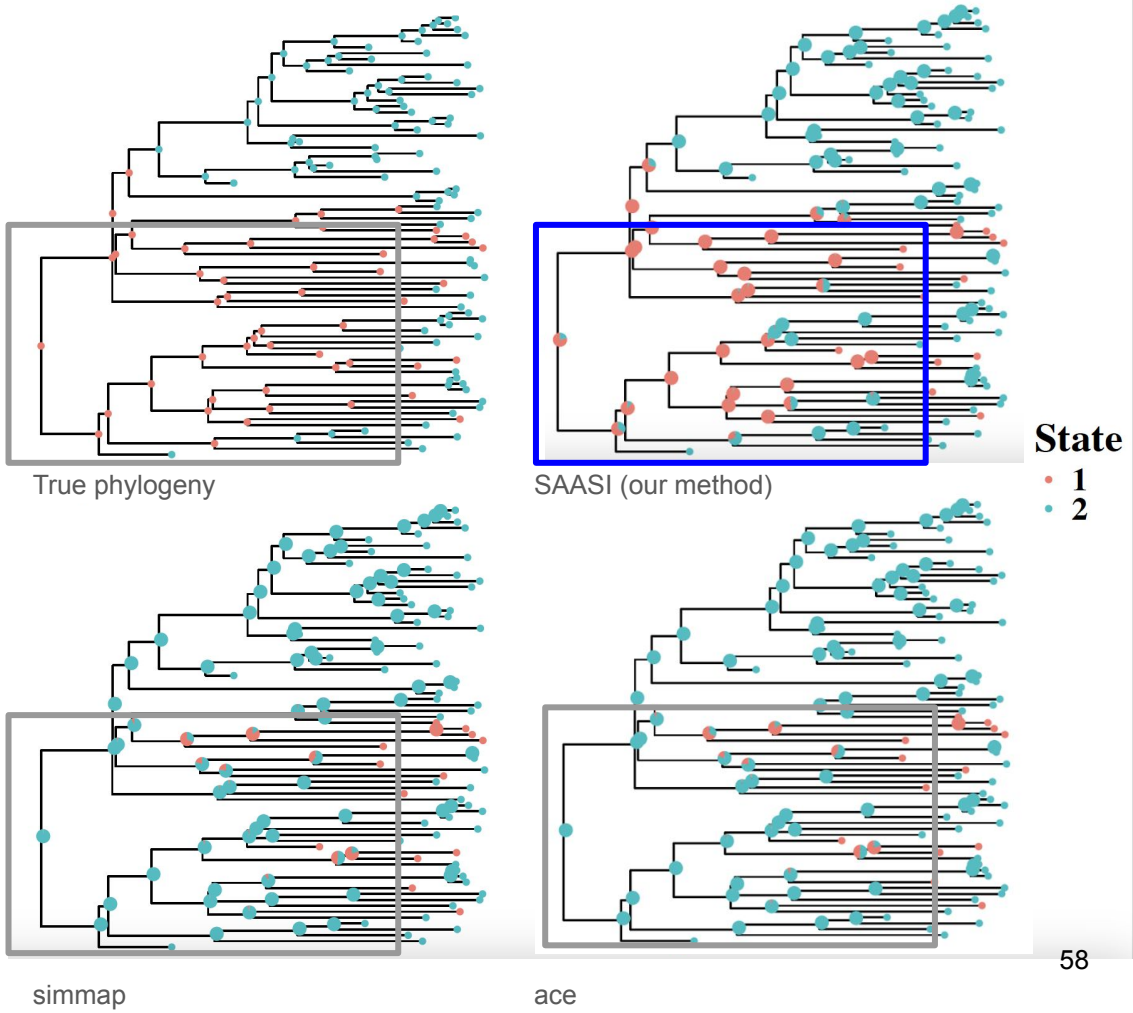
SAASI corrects for this via E_i and $D_{e,i}$.



Example with two states

- State 1 (red) **50** times less sampling than State 2 (cyan)
- Accuracy: fraction of nodes with correct inferred location
- ace & simmap: affected by sampling; 58% accuracy
- SAASI infers ancestral states more accurately (88% in this example).

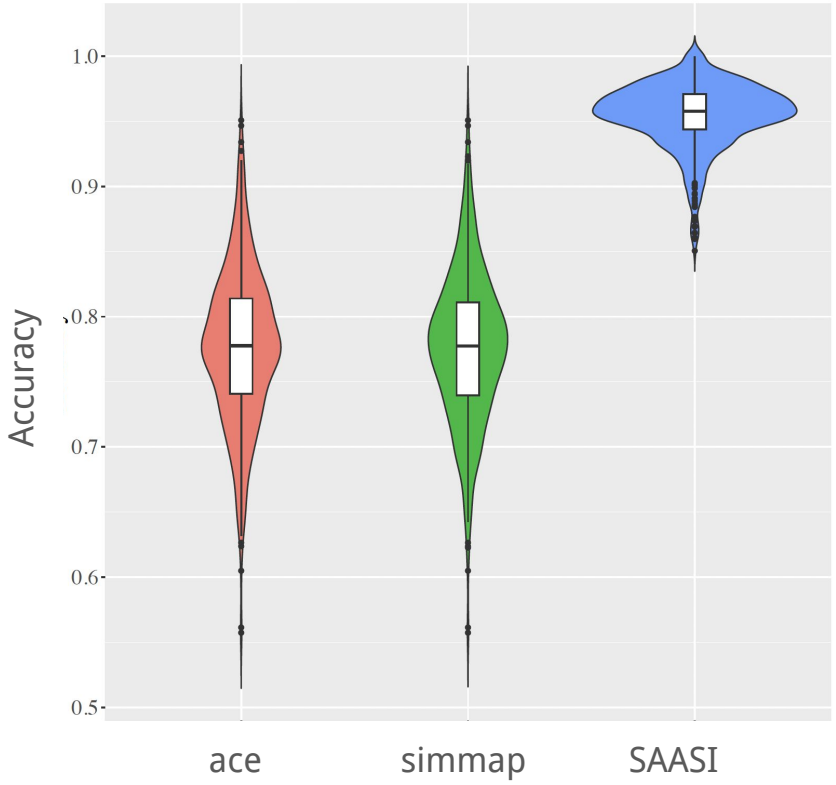
State 2 is over-represented in the tree: ace and simmap wrongly place too many nodes in state 2.



Accuracy over 500 simulated trees

- Simulation: 500 trees. 2 states. Known states at all internal nodes.
- Assume State 1 has 100 times less sampling than State 2.
- Compare SAASI to two standard methods (ace, simmap)
- Accuracy is measured as the overall accuracy:

  fraction of internal nodes whose internal node state is assigned correctly (max probability state)



H5N1 HA demonstration

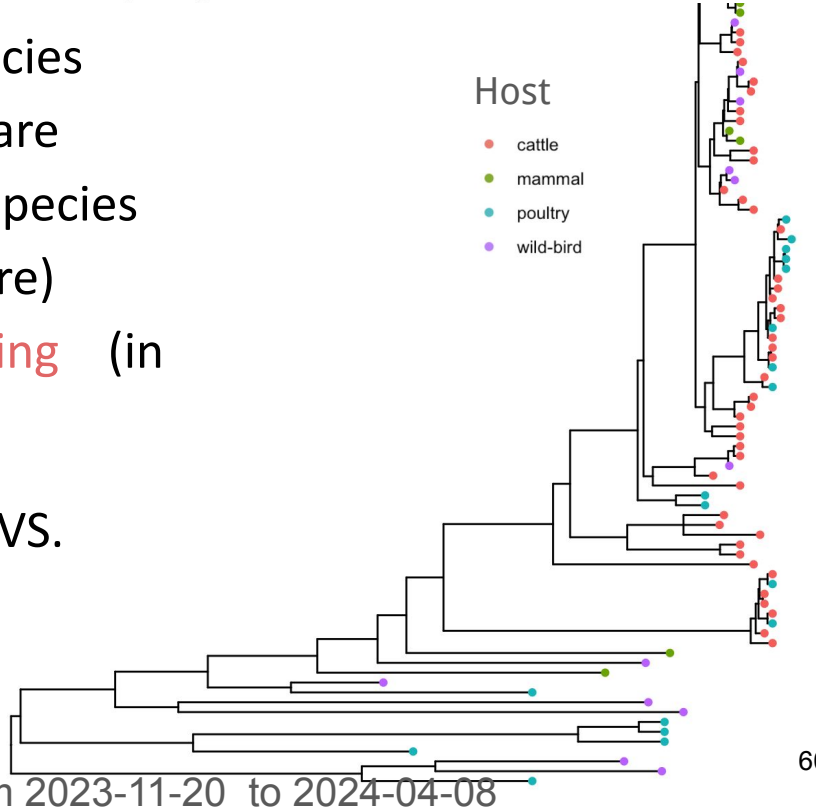
Emergence and interstate spread of highly pathogenic avian influenza A(H5N1) in dairy cattle

Thao-Quyen Nguyen, Carl Hutter, Alexey Markin, Megan Thomas, Kristina Lantz, Mary Lea Killian, Garrett M. Janzen, Sriram Vijendran, Sanket Wagle, Blake Inderski, Drew R. Magstadt, Ganwu Li, Diego G. Diel, Elisha Anna Frye, Kiril M. Dimitrov, Amy K. Swinford, Alexis C. Thompson, Kevin R. Snevik, David L. Suarez, Erica Spackman, Steven M. Lakin, Sara C. Ahola, Kammy R. Johnson, Amy L. Baker, Suelee Robbe-Austerman, Mia Kim Torchetti, Tavis K. Anderson

doi: https://doi.org/10.1101/2024.05.01.591751

- 104 taxa, HA segment, multiple host species
- We suspect that infections in wild birds are undersampled compared to other host species (more accessible, eg domestic, agriculture)
- We want to examine the effect of sampling (in wild birds in particular).

Data: Nguyen *et al* Fig. S2. Used BEAST + BSSVS.



https://github.com//andersen-lab//avian-influenza

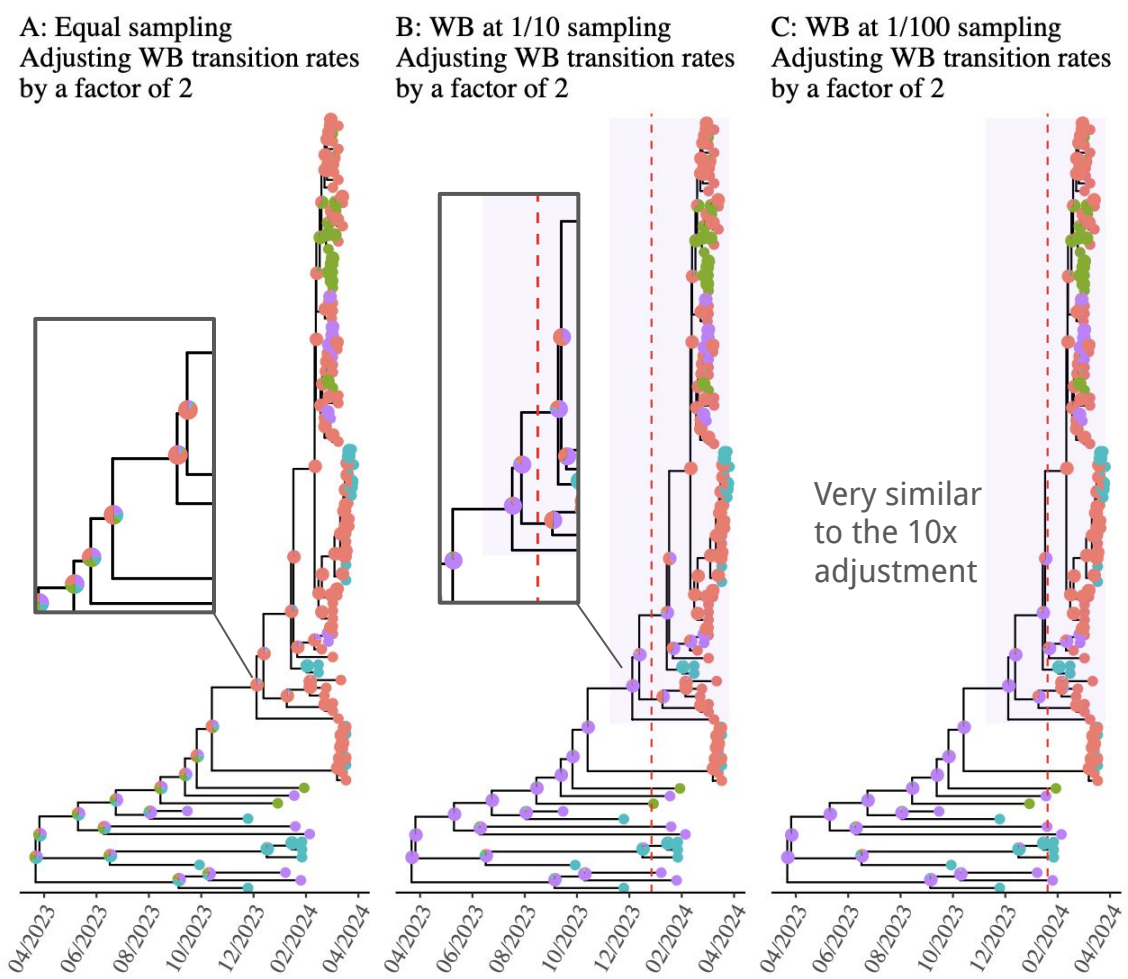
Effect of sampling in wild birds

No sampling adjustment: same result as ace, Nguyen et al:

- one introduction to cattle
- Dec 2023-Jan 2024.

With sampling adjustment:

- Multiple introductions
- late Jan-early Feb 2024.



Method inputs are a challenge

The **bad news**: node state probabilities are defined using differential equations that use state-dependent branching, extinction and sampling rates, and inter-state transition rates. These must be estimated.

For now this is done separately,
but in principle it can be done in the same model

$$\frac{d}{d\tau}D_{e,i}(\tau) = -\left(\sum_{j,k}\lambda_{i,j,k} + \mu_i + \psi_i + \sum_{j\neq i} q_{i,j}\right) D_{e,i}(\tau) + \sum_{j,k} 2\lambda_{i,j,k} D_{e,j}(\tau) E_k(\tau) + \sum_{j\neq i} q_{i,j} D_{e,j}(\tau).$$

The **good news**: results are quite robust to mis-specification of these parameters. We use ace for the transition rates, and use simulation tests to adjust ace's rates.

Method implementation: saasi R package

Inputs:

- A phylogenetic tree; tip states
- Sampling rates (if not known, SAASI works as a sensitivity analysis)

Intermediate (SAASI has tools to estimate):

- Parameter estimates (birth rate, death rate, transition rate matrix)

Output

- The phylogenetic tree with estimated node states (vector of probabilities)

Practicalities

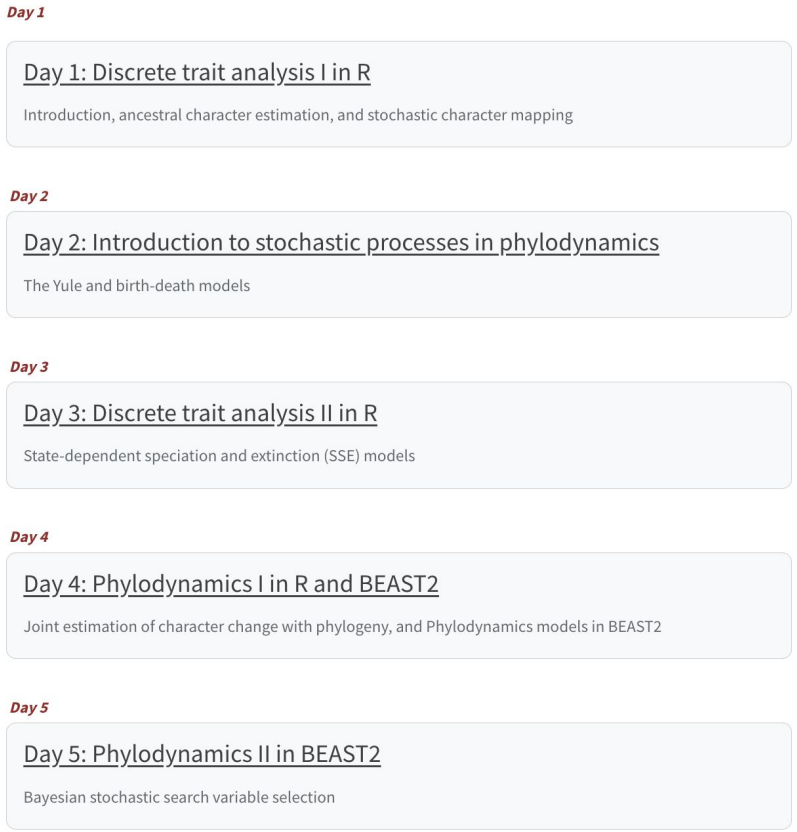
- Time is linear in the number of tips. With 2 states, **100,000 tips**: **1 hour**.
- R package: https://github.com/MAGPIE-SFU/saasi

Phylogeography workshop (online)

https://alexbeams.github.io/phylogeography_workshop/

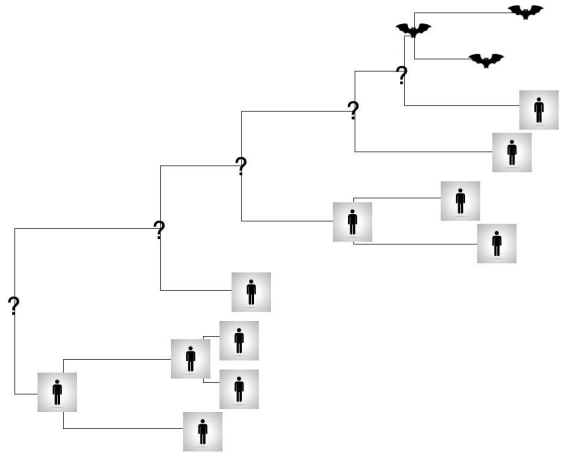
- practical phylogeography
- in-depth theoretical explanations
- the birth-death models
- forward and backward equations
- R exercises that demonstrate how to actually perform phylogeographic reconstructions in ACE, SAASI
- Stochastic character mapping
- Available online
- Video lectures coming
- Slides and R content available now

Thank you to Alex Beams in my group



Quiz: Sampling and phylogeography

1. In the Ebola toy example (right), why will ACE place most of the ? nodes in humans?
2. Suppose the sampling rate for state *i* is very small. Is *Ei(t)* large or small?
3. How does that change our belief that an ancestor node is in state *i*?
4. Why do you think that sampling wild birds 1/10 as much as others, and 1/100th as much, leads to very similar results?



Quiz answers

1. In the Ebola toy example (right), why will ACE place most of the ? nodes in humans?

   A: *Most tips are human, and the model has no term for how likely each state is to be sampled. Having fewer state changes is usually favoured, leading to human at the ? nodes*

2. Suppose the sampling rate for state *i* is very small. Is *Ei(t)* large or small?

   A: *Larger - it's more likely that the lineage has no sampled descendants (E)*

3. How does that change our belief that an ancestor node is in state *i*?

   A: *This would increase our belief that an ancestor is in state i without having been sampled*

4. Why do you think that sampling wild birds 1/10 as much as others, and 1/100th as much, leads to very similar results?

   A: *Intuitively, the 1/10 value is enough to remove the constraints on the ancestry estimates from not having tips in wild birds. With those constraints removed, it can attribute ancestry to wild birds.*

Part 4: Reconstructing transmission trees with sequence data

What is “reconstructing transmission” and why do it?

- infer who infected whom (with uncertainty)
- estimate **how fast** transmission happens
- estimate how many are **missing**, and where in the tree?
- estimate **patterns** in **who infected whom**: what is associated with transmission?
- infer **transmission locations and routes** - geographic, other

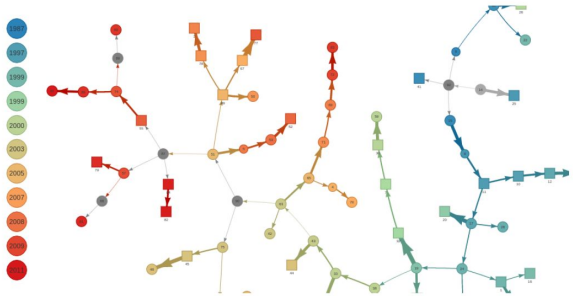
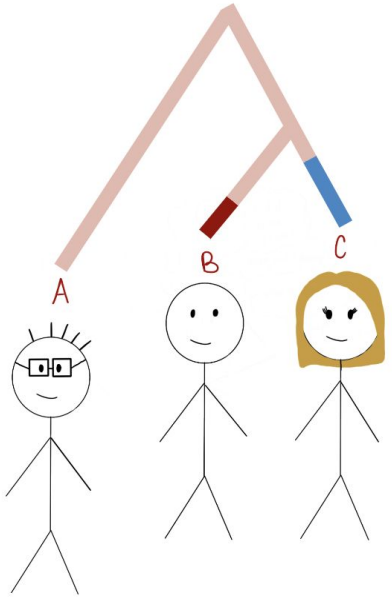


Illustration of who infected whom with uncertainty and unsampled cases

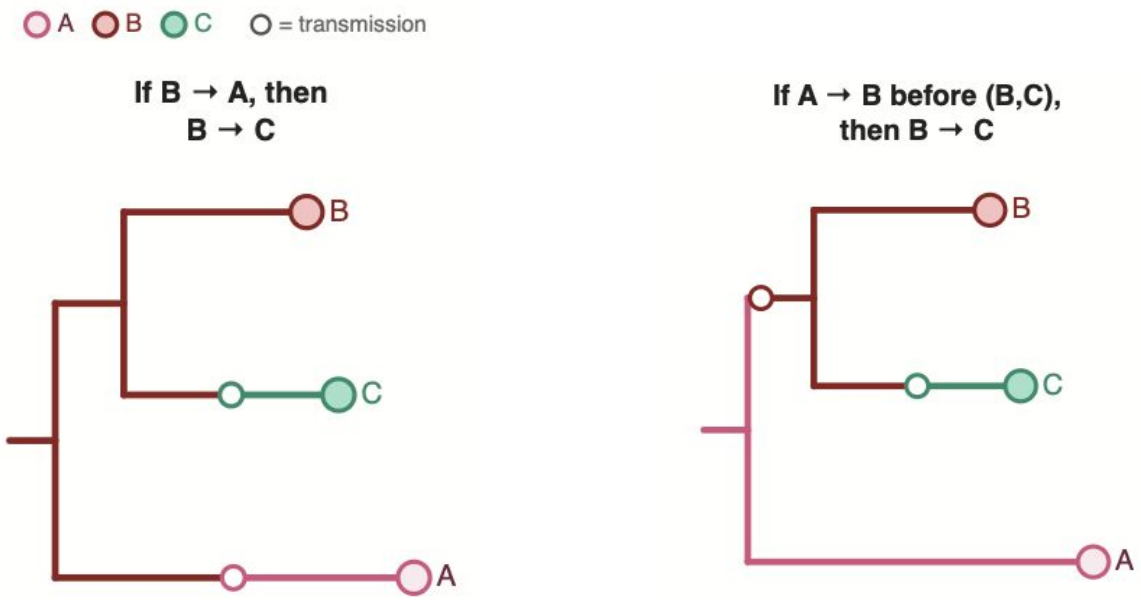
Estimating who infected whom

- Challenges:
  - phylogenetic uncertainty
  - transmission uncertainty
  - within-host diversity
  - unsampled individuals
- Previous methods: outbreaker2 (pairwise method); TransPhylo (fixed tree); phybreak (no unsampled cases); BEASTLIER (no unsampled cases)
- There are additional challenges as well



A infects B and C.
This is consistent with this (A, (B,C)) phylogeny because multiple edges are in A at the same time (A's infection is diverse)

Phylogenetic trees constrain transmission trees (somewhat)



Colour represents which person the edge segment is in. Each segment is in one person. Colours are continuous.

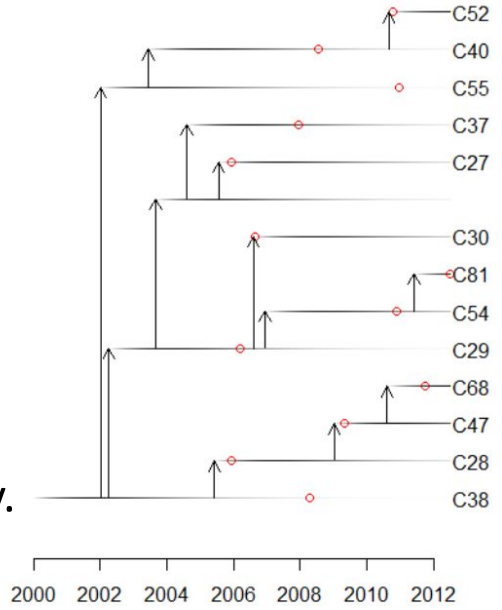
Previous approach: TransPhylo

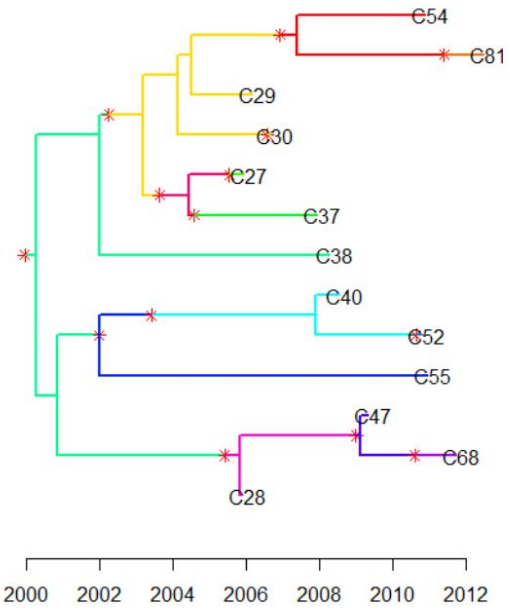
Colour represents the transmission tree (augmentation)

Unsampled individuals are added to edges as individual infections with start and end times.

Use MCMC: compute likelihood, accept or reject, continue.

The fixed phylogeny was a major limitation: low diversity gives phylogenetic uncertainty.



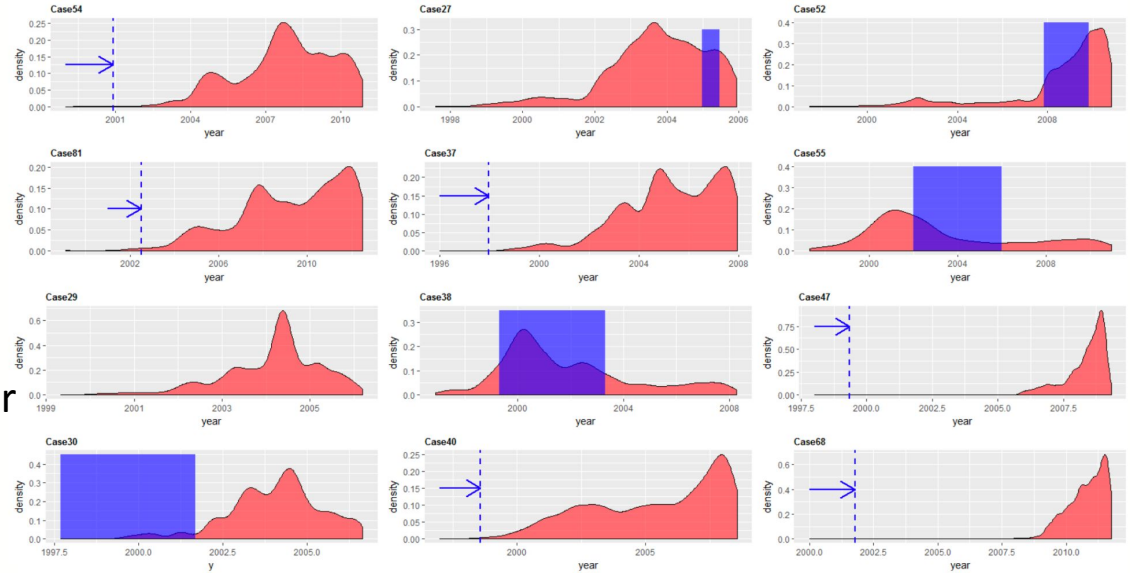


TB transmission inferences with TransPhylo

Example 1: *where* were individuals infected?

Norway: data from genomics suggested recent transmission in Norway.

Connect *when* to *where*: infected after arrival in Norway?



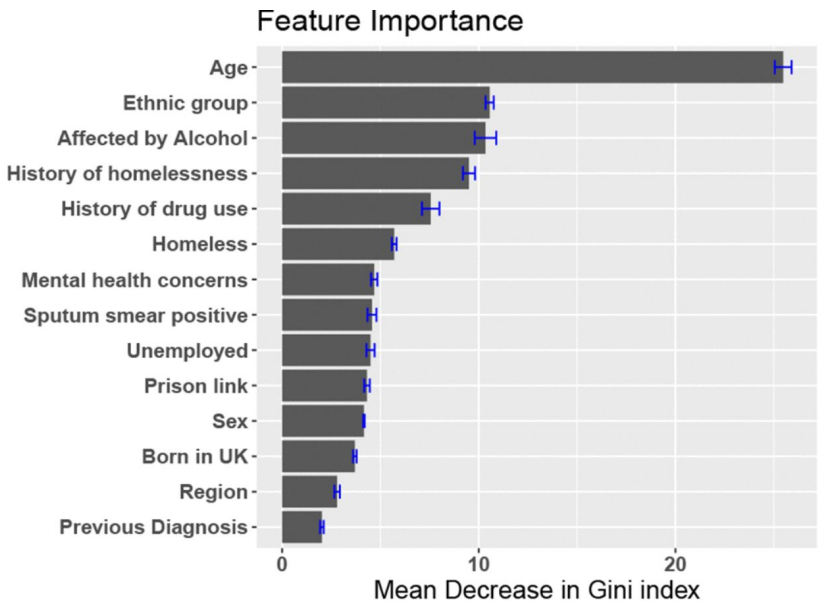
Red: Posterior time of infection. Blue: arrival in Norway.

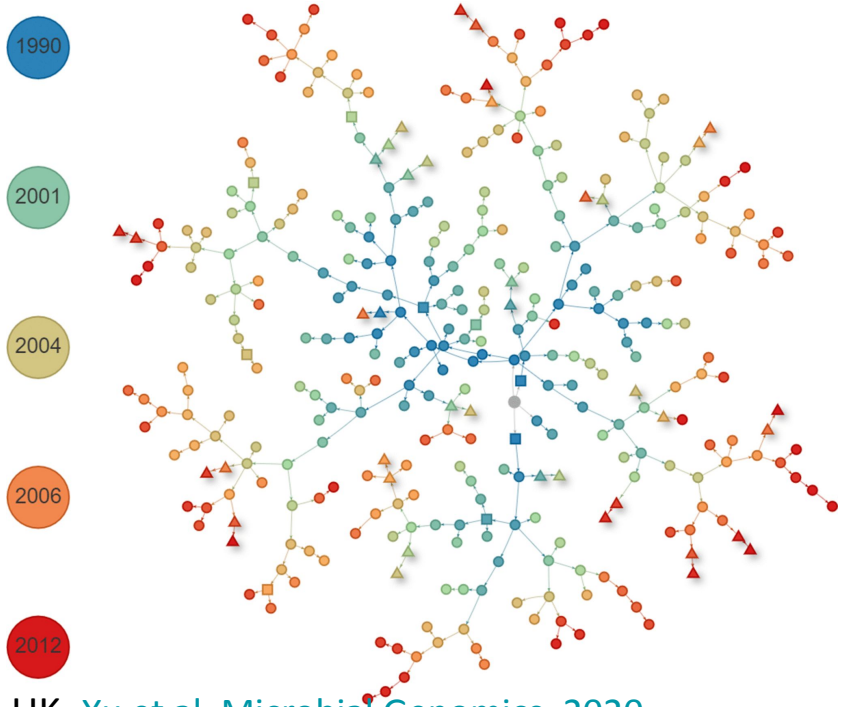
Ayabina et al, Microbial Genomics, 2018

Collaboration with V. Eldholm

TB transmission inference with TransPhylo

What is associated with being a plausible transmitter?





INH-R outbreak, London, UK. Xu et al, Microbial Genomics, 2020

BREATH: Bayesian Reconstruction and Evolutionary Analysis of Transmission Histories

Simultaneous Bayesian inference of:

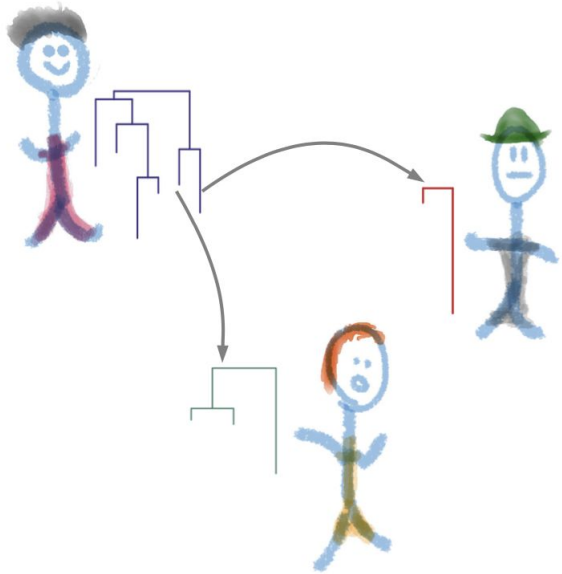
1. Timed phylogenetic tree

2. Transmission tree (who infected whom and when)

Input: sequences, dates, timing information

Output: posterior collection of transmission trees and timed phylogenies.

Paper: Colijn, Hall and Bouckaert, *MBE* 2026



Method ingredients

- The likelihood of the transmission tree given what we know about the epidemiology -- BREATH model
- The likelihood of the phylogenetic tree given the transmission tree (use independent coalescent trees within hosts)
- The likelihood of the sequence data given the phylogenetic tree (use the established tools in BEAST2)
- MCMC: propose, compute likelihoods and priors; accept/reject, continue.

BREATH works in BEAST2, a leading phylogenetic reconstruction package.

BREATH is a Bayesian method, with posterior given by:

$$P(G, T, \theta, N_e g, w | D, \tau_s) \propto P(D|G, w)P(G|T, N_e g, \tau_s)P(T, \tau_s|\theta)Pr(\theta)Pr(N_e g)Pr(w)$$

Built from the usual conditional probability:
P(A|B)P(B) = P(B|A)P(A)

G: genealogy

T: transmission tree

D: data

$\boldsymbol{\tau}$: sampling times

$N_e g$: within-host coalescent parameter

$\boldsymbol{\theta}$: epidemiological parameters

w: molecular clock model

The key ingredients and the missing piece

$$P(D|G,w)P(G|T,N_eg,\tau_s)P(T,\tau_s|\theta)Pr(\theta)Pr(N_eg)Pr(w)$$

Known: Data given phylogeny. Standard in phylogenetics.

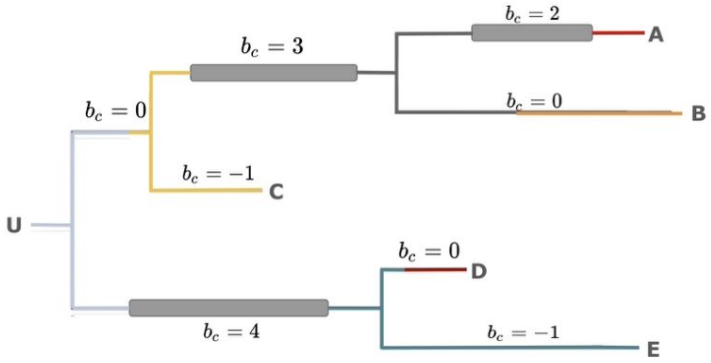
Easy: Phylogeny given transmission tree. Independent evolution in each individual host (colour). Chop tree up into small simple pieces.

Priors:
Known, easily defined.

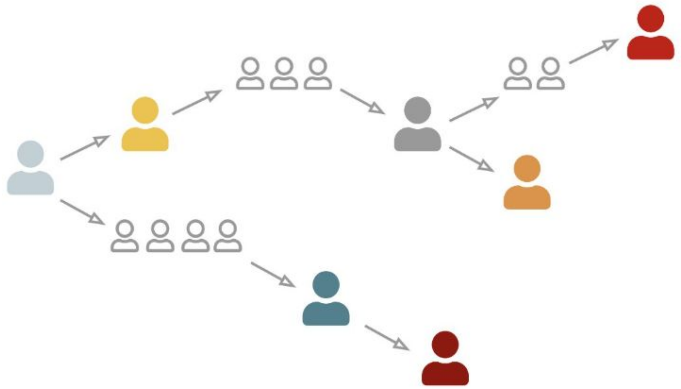
Missing piece: Transmission tree given epidemiological knowledge.
A main challenge: allow unsampled cases, in real time.
This is what I'll describe next.

How BREATH relates transmission to phylogeny

We associate each phylogenetic tree with a transmission tree. We allow any number of unsampled cases in cryptic transmission chains.



A. Phylogenetic tree with colouring and block counts



B. Corresponding transmission tree

The stochastic process: transmission and sampling

Individual cases:

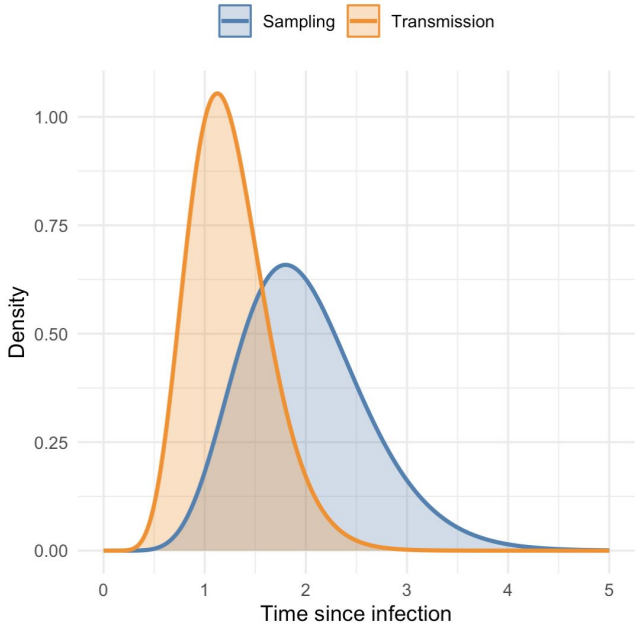
There is a hazard of transmission and a hazard of sampling.

We account for the end time (last possible time of sampling)

Unsampled transmission chains:

We account for cases being unsampled and infecting one other who is ancestral to the sample.

We account for the duration of these “cryptic” transmission chains.

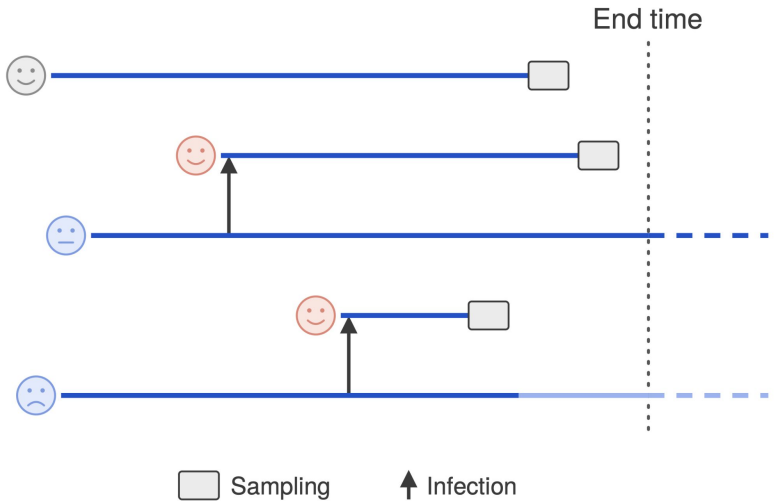


We account for “right truncation”: things had to happen before the present (or before the last observation time)

We have no knowledge of this individual at all (note, contrast with censoring).

We have indirect knowledge of these ones (blue), because they infected someone who is sampled

We are more likely to see those who are sampled or who infect others *sooner.*



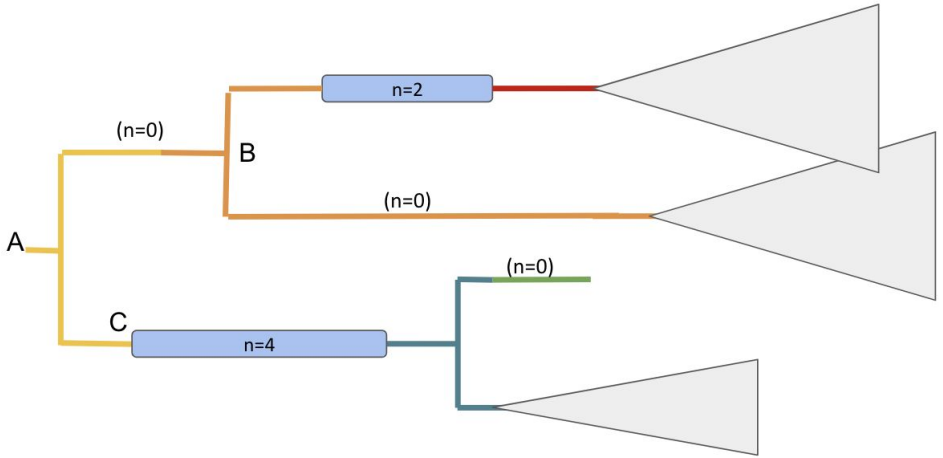
We account for this in the probability that the event happens before or at a given time. This is an established approach in survival analysis.

Transmission tree likelihood

B's likelihood depends on B's time of infec
when A infected B.

T_x is the tree descending from x.

Let c(x) be the "children" of x (the onward
descendants).



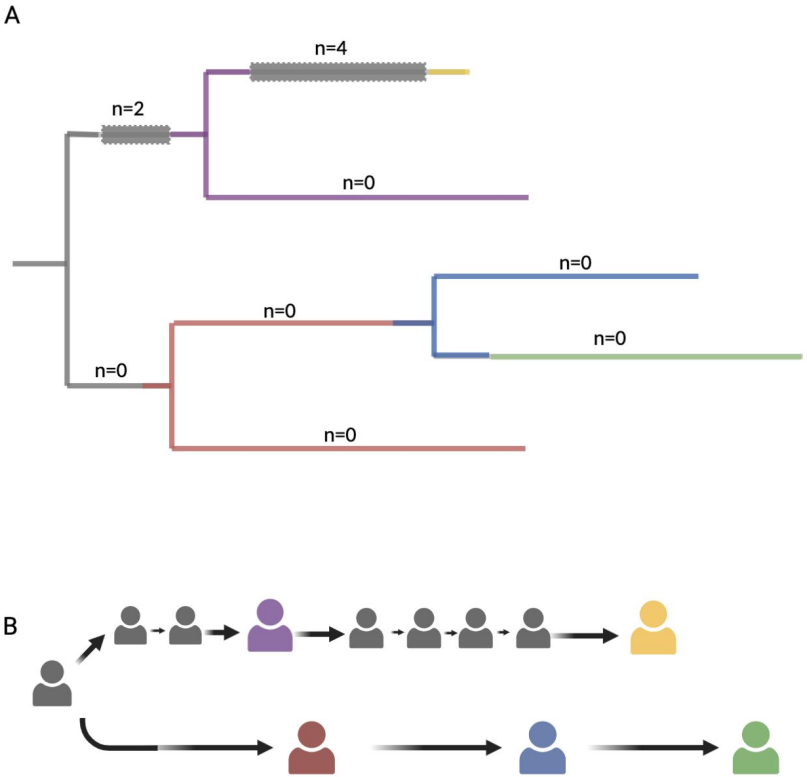
$$L(A, B, C, \ldots) = L(A)\ L(T_B|A)L(T_C|A) = L(A)\ \ L(B|A)L(T_{c(B)})\ \ L(C|A)L(T_{c(C)})$$

Get a product: one term for each case, one term for each unsampled chain

Building the transmission likelihood

- There are 3 kinds of people in the tree: sampled individuals, individual unsampled hosts, chains of (unsampled) transmission
- Each edge has an number *n*.
- *n* represents the number of *additional* hosts on the edge, in a chain of unsampled transmission.
- Outside the blocks, edges and nodes have a colour indicating the host
- Nodes of the phylogeny have to be inside individuals (not chains of unsampled transmission).

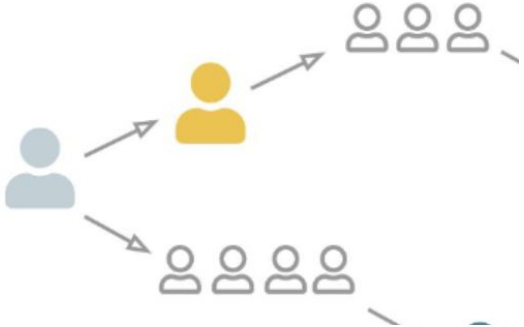
An annotated phylogeny (A), and the associated transmission tree (B)



Ingredients to make the transmission likelihood

1. Likelihood for an individual case

This blue individual infects two others and does not get sampled. He is infected at time *t_blue* and infects the others at times *t_yellow* and *t_block*.



2. Likelihood for an unsampled chain of transmission

This chain of 4 unsampled cases starts at time *t_block*. It ends when the 4th case infects someone who eventually is sampled (dark green), at time *t_green.*



Sampling and transmission contributions to the likelihood

We model two kinds of events: sampling, and transmission to another individual.

Each has an intensity function, h^s and h^{tr} respectively, with

$$h(t) = \lim_{dt \to 0} \frac{P(\text{event occurs in time } [t, t+dt])}{dt}.$$

Let $S(t) = \exp\left\{-\int_0^t h(s)ds\right\}$.

The likelihood of k events at times $t_1 < t_2 < \ldots < t_k$ with intensity $h(t)$, in an interval from 0 to t, is the probability that events do not happen except at times t_j:

$$L = \exp\left\{-\int_0^t h(s)ds\right\} \prod_{j=1}^{k} h(t_j) = S(t) \prod_{j=1}^{k} h(t_j)$$

except at times t_j

nothing happens in *all* the little time intervals

Individual cases' likelihood

$$t_s^k = \begin{cases} \text{time from infection to sampling (if sampled)} \\ \text{time from infection to study end, if not sampled} \end{cases}$$

t_e^k = time from infection to cure, or end of study

T = time from infection to end of study – right truncation time

1-$S^E(T)$ conditions on transmission or sampling happening by the end time

The likelihood for an individual case k:

$$L(k|\theta, \tau_i^k) = \underbrace{S^s(t_s^k) h^s(t_s^k)^{1_{k \text{ samp}}}}_{\text{sampling happens, only at the sample time}} \underbrace{S^{tr}(t_e^k) \prod_{j:k\to j} h^{tr}(t_i^j)}_{\text{transmission happens, only at the specified time}} \frac{1}{1 - S^E(T)}$$

Chains of unsampled transmission: how do they end?

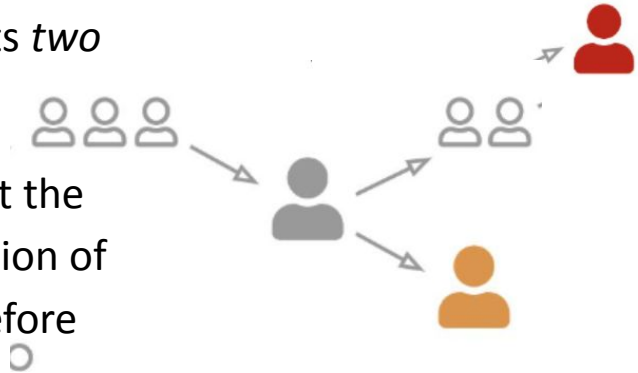
Chains of unsampled transmission continue while

- a case remains unsampled, and
- the case infects another who is “once-ancestral to the sample”



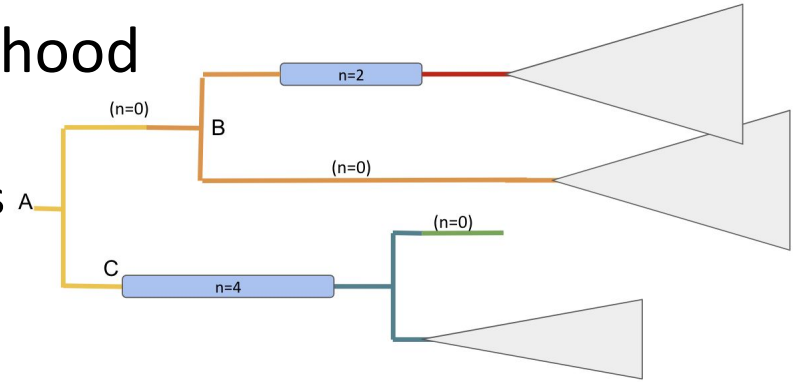
Blue above: the chain ends because the blue case is sampled.

Grey: the chain ends at a phylogenetic node. The grey case infects *two* others who are ancestral to the sample.



We use branching process theory to work out the probability that the chain continues. Given *n* cases, we know the density of the duration of the chain. Finally, we condition on only seeing chains that end before the end time.

This completes the transmission likelihood



We have contributions from the individual cases (with their times of sampling and infecting others) and the chains of transmission.

$$L(A, B, C, ...) = L(A)\ L(T_B|A)L(T_C|A) = L(A)\ \ L(B|A)L(T_{c(B)})\ \ L(C|A)L(T_{c(C)})$$

This was the main missing piece. We can now compute all terms in the posterior (numerator), so we can sample phylogenetic trees and transmission trees with BEAST2.

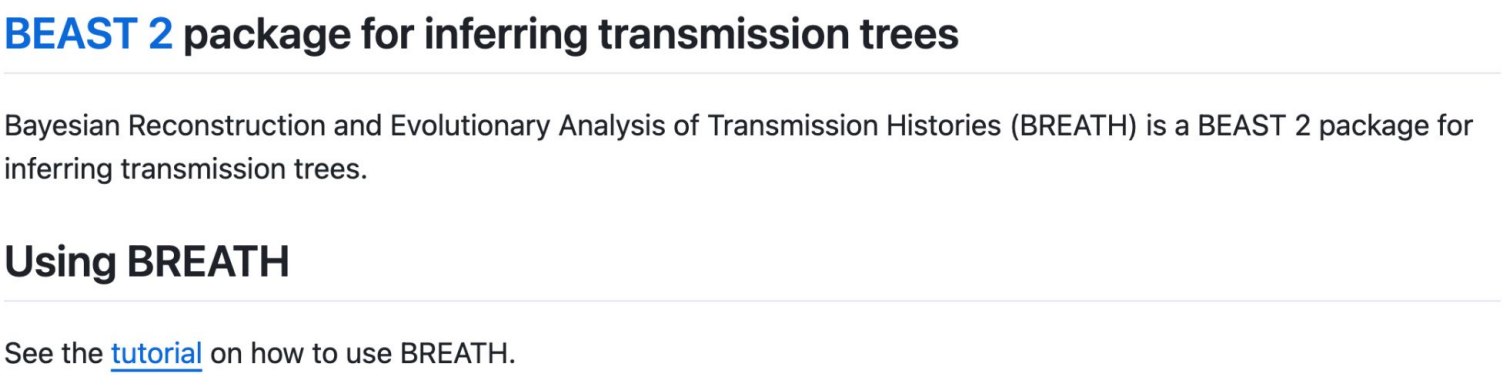
$$P(D|G, w)P(G|T, N_e g, \tau_s)P(T, \tau_s|\theta)Pr(\theta)Pr(N_e g)Pr(w)$$

Method implementation: BEAST2

BEAST2 implementation by Remco Bouckaert: the *BREATH* package.
https://github.com/rbouckaert/BREATH

This allows us to *simultaneously* estimate

(1) the phylogenetic tree for the sequences
(2) the transmission tree: who infected whom, and when,
(3) including where there are chains of unsampled transmission.



How we test the method: Simulate, and see if it is right

Simulation model:

We simulate transmission and sampling with intensities *h* following the model.

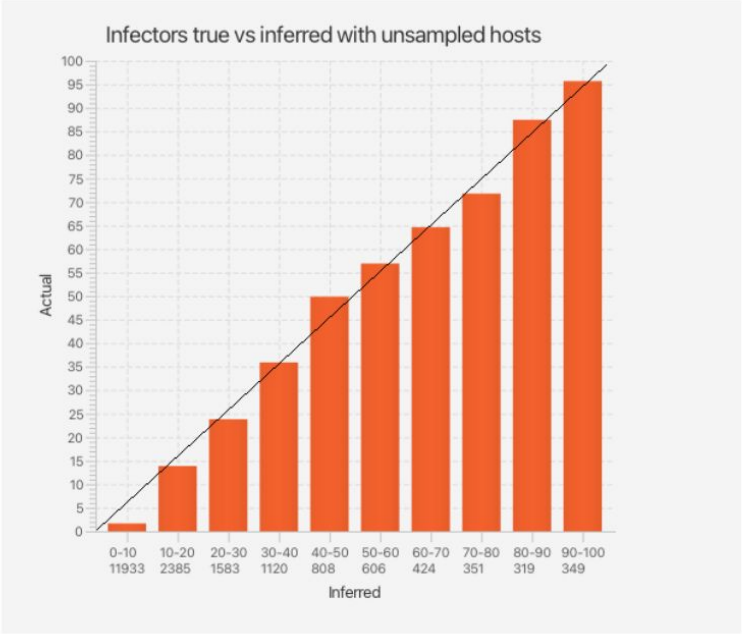
We simulate within-host phylogenies with a constant-rate coalescent.

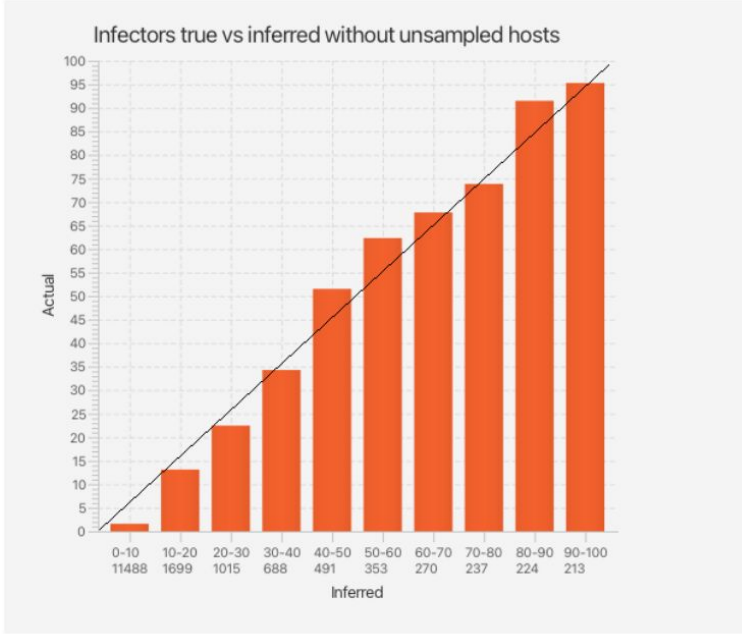
This allows us to check model performance.

We use a “well calibrated simulation study”: compare our posterior probabilities to the fraction of time an event happened (in the simulation).

“Well-calibrated simulation test”

Suppose our model says that A infected B with probability 0.4.
Now think of all the events our model says happen with probability 0.4.
If the model is working, a fraction 0.4 of these events should have happened.





TB outbreak application

Comparative Study > PLoS Med. 2013;10(2):e1001387. doi: 10.1371/journal.pmed.1001387. Epub 2013 Feb 12.

Whole genome sequencing versus traditional genotyping for investigation of a Mycobacterium tuberculosis outbreak: a longitudinal molecular epidemiological study

Andreas Roetzer [1], Roland Diel, Thomas A Kohl, Christian Rückert, Ulrich Nübel, Jochen Blom, Thierry Wirth, Sebastian Jaenicke, Sieglinde Schuback, Sabine Rüsch-Gerdes, Philip Supply, Jörn Kalinowski, Stefan Niemann

Data:

TB outbreak of 86 cases in Hamburg, Germany.

Active case finding and passive surveillance took place.

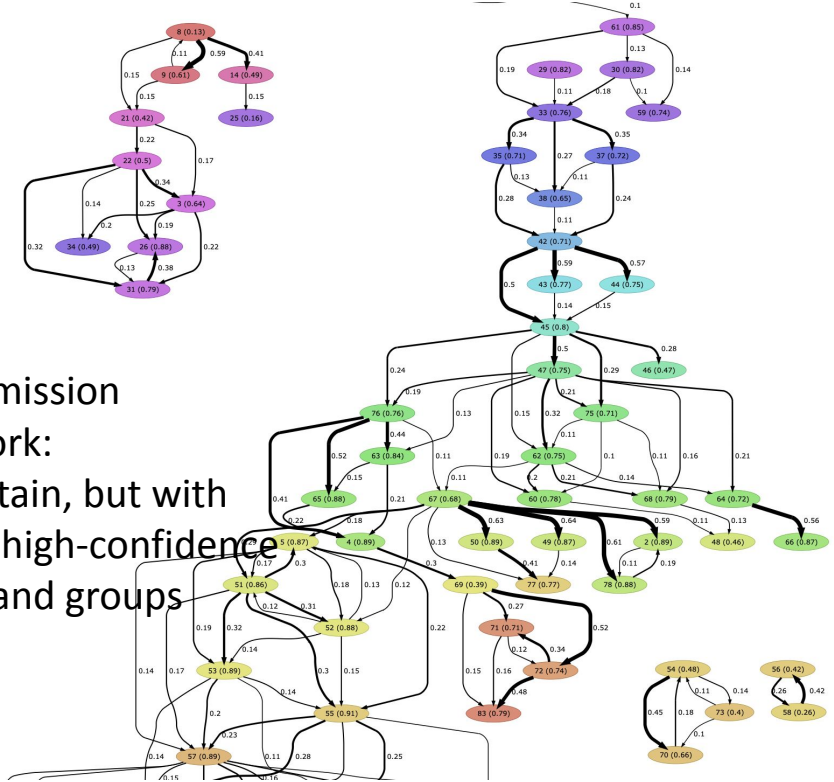
Genomic data (SNPs) and times are publicly available.

In an attempt to stop the outbreak, public health closed a bar in the town where most cases were identified. More cases emerged, though, several years later.

Our previous method (TransPhylo; Didelot et al 2017) uses this data (of Roetzer et al, 2013)

TB outbreak: 86 cases in Hamburg, Germany (Roetzer et al 2013)

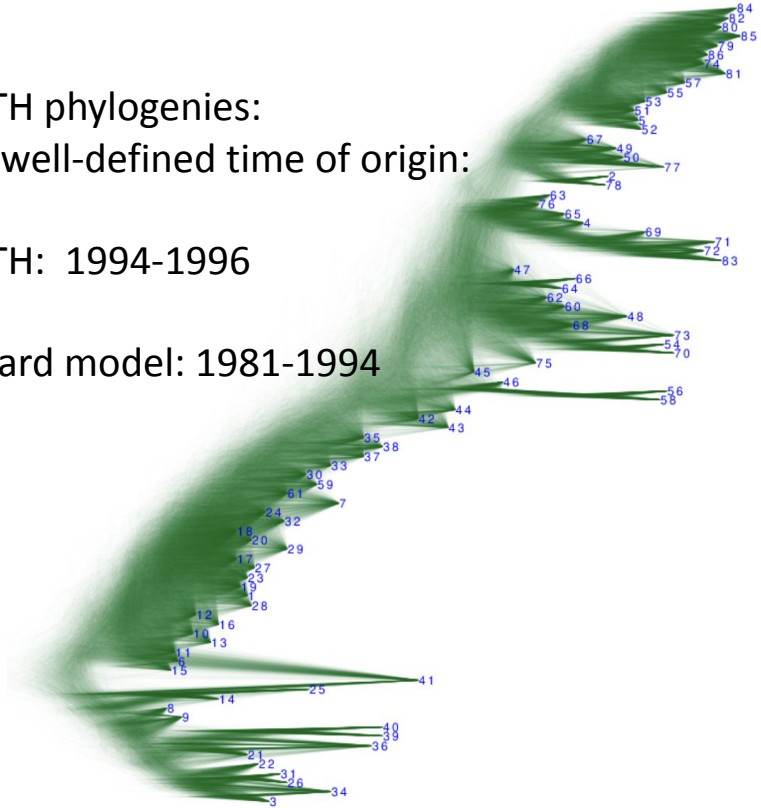
Transmission network:
uncertain, but with some high-confidence pairs and groups



BREATH phylogenies:
more well-defined time of origin:

BREATH: 1994-1996

Standard model: 1981-1994



Quiz: Transmission Reconstruction

1. Why can’t we simply read who infected whom from the branching pattern of the phylogeny?
2. In the Hamburg TB outbreak, new cases appeared years after the bar was closed. What could explain this?
3. What could a method like BREATH tell public health that the case list alone could not?

Quiz Answers

1. Why can't we simply read who infected whom from the branching pattern of the phylogeny?

   A: *They are different objects. Multiple transmission trees are possible on the same phylogenetic tree.*

2. In the Hamburg TB outbreak, new cases appeared years after the bar was closed. What could explain this?

   A: *Unsampled chains of transmission, either in Hamburg or beyond; also TB has a long latency period which could be a factor*

3. What could a method like BREATH tell public health that the case list alone could not?

   A: *BREATH can estimate where unsampled chains are likely and who probably infected whom, which helps target contact tracing and screening. It can estimate how many lineages were circulating during the two phases of the outbreak.*

Summary: Phylogenetics can be very useful in epidemiology!

Question	Method	Inputs	Key assumptions	Related methods
Phylogenetics: What is it? When did it start?	Timed phylogenetic tree	Dated sequences, reference sequences	Molecular clock: substitutions accumulate at a steady rate	Least squares dating, Bayesian dating (BEAST2)
Phylodynamics: How fast is it spreading (R, R(t))?	Birth–death–sampling model	Timed tree or sequences; priors (e.g. infection duration)	Well-mixed population; constant or piecewise rates; sampling at rate ψ	Coalescent (Ne(t)), skyline plots, deep learning (PhyloCNN)
Phylogeography: Where was the pathogen in the past?	Discrete trait reconstruction (ace)	Tree, tip locations	Location evolves as a Markov chain on the tree; equal sampling across locations	Bayesian phylogeography (BEAST), stochastic mapping (simmap)
…when sampling differs by location?	SAASI	Tree, tip locations, sampling rates (or a range)	Multi-type birth–death process; rates estimated separately	SSE models (BiSSE)
Transmission: Who infected whom, and when?	BREATH	Sequences, dates, timing information	Within-host coalescent; transmission and sampling hazards; unsampled chains	TransPhylo

Thank you

Phylogeography: Alex Beams (workshop)

SAASI: Yexuan Song, Ivan Gill, Ailene MacPherson

BREATH: Matthew Hall, Remco Bouckaert

Ebola: Placide Mbala, Eddy Kinganda, Andrew Rambaut, ARTIC 2.0, Renny Doig

VIASM for hosting us!

Olivier Gascuel and Nguyễn Hồng Anh and other organizers